

MLX2

Technology & Research Review

V2 · MEASURED · FINAL

What happened to every recommendation of the first review once it met an Apple GPU: which ideas were built, which were measured and rejected, what the sweep found that the review could not see from source, and what the final A/B runs on the integration tree showed.

Foundations	A1 PENDING routes rerun twice dossier not written	A2 ADOPTED p95 -15% / -23% identical-code A/B	A3 FIXED 55.47% error fixed serving row fixed	
Policy & kernels	B1 REJECTED planner ~ 0.5% below the 3% bar	B1' ADOPTED 8K TTFT -7% bit-identical	B2 OPT-IN stage one 0.83× d8 synthetic only	B3 OPT-IN CPU replay only trace A/B not run
Algorithms	C1 OPT-IN exact, +5-18% E[acc] GPU A/B not run	C2 DEFERRED +2.4...9.0% bound approximate · not built	C3 REJECTED ≈ 1% prize, inexact rejected	
Research bets	D1 PENDING risk refuted not scheduled	D2 DEFERRED not built controls defined		

REVIEW DATE	7 October 2026, with the final evidence of the 6 October sweep
SOURCE	main at 43d561e38, baseline origin/main 187c6aa23; v1 pinned 008eb011fe49
PUBLISHED	GitHub pierre427/mlx2 8359344 (13 signed topic commits) · private main 43d561e38
HARDWARE	Apple M5 Max, 128 GB; GPU campaigns through the real HTTP serving path
EVIDENCE	17 GPU campaigns (baseline, intermediate, final), research microbenchmarks, exhaustive CPU oracles, 183 sweep commits

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BEFORE YOU START

How to read v2

v1 was a source audit with CPU probes on Linux and no Apple GPU. v2 keeps its structure and its sources, and adds what real M5 Max runs, a ten-lane code sweep and the final A/B on the integration tree established.

Evidence labels

M5 GPU · MEASURED

CPU · REAL CODE

REPOSITORY-RECORDED

PROPOSED

PENDING

LABEL	MEANING
M5 GPU · MEASURED	Observed on the M5 Max GPU in a campaign or microbenchmark committed with this review. Two or three reps, rotated order, swap checked. A/B evidence: not qualification, not a controlled performance run.
CPU · REAL CODE	The repository's own code executed on the CPU: exhaustive oracles, closed-loop policy simulators, serving-engine repros on tiny models. Exact about logic; says nothing about GPU time.
REPOSITORY-RECORDED	A result recorded earlier in the repository or by a sweep lane, not re-run for v2.
PROPOSED	A design or estimate. Not measured.
PENDING	The deciding run has not happened (chapter 11 lists each one).

Verdicts

ADOPTED · DEFAULT

OPT-IN

FIXED

REJECTED

PENDING

DEFERRED

Adopted means selected by default in source, which is not the same as qualified. **Opt-in** means implemented behind an explicit switch with default behaviour unchanged. **Rejected** means measured and not worth building or selecting. **Fixed** means a defect v1 or the sweep found is corrected in source. **Pending** means the deciding measurement has not run. **Deferred** means consciously queued behind other work.

Status blocks

Each of v1's ten highest-value decisions and its eleven roadmap items (A1-A3, B1-B3, C1-C3, D1-D2) gets one block, plus B1', the 6-bit NAX kernel that B1's measurement led to. Each block says: was it built (commit or branch), was it tried (evidence file), what was measured, the verdict, and what v1 got right or missed. Citations jump to the register: **C**, **R** and **P** keep v1's identifiers; **S** are sweep commits; **E** are evidence files.

FOUR KINDS OF CORRECTNESS, UNCHANGED FROM V1

State and protocol correctness, numerical equivalence to the ordinary route, sampling-law exactness, and approximate quality are different claims. Every number below is tied to one of them, and to one of the states *implemented*, *qualified*, *selected* and *observed-used*, which remain distinct.

CHAPTER 1

Executive summary

Most of v1 held up. The parts that did not were the size of the prizes, and the GPU showed problems that a source audit could not. The final runs show which fixes held.

5 / 11

v1 roadmap items built in some form (commit or branch)

2 + B1'

adopted by default or fixed; plus 6-bit NAX, found by B1

2

measured on the GPU and rejected: B1 planner (0.5%), C3 GDN (≈1%)

5

opt-in or still pending a deciding run

FINAL A/B ON THE INTEGRATION TREE

46-51 → 104-111

27B native MTP, B4 aggregate tok/s; now 1.5× ordinary (SS-1 cohort fix) [E15]

6.9-9.5 → 0.44-0.47 s

Flash-Next served MTP: longest pause of a decoding neighbour beside a 16K prefill [E17]

-7% / -8%

35B 8K / 16K TTFT from 6-bit NAX (B1'), bit-identical output [E18]

-15% / -23%

neighbour gap p95, cost model vs running max (MTP / ordinary), identical code [E16]

v1 argued that MLX2 gains more from making its advanced paths measurable, numerically coherent and cheap to combine than from new feature names. The GPU runs agree, and sharpen it: the largest measured effects came from scheduling and serving seams (a 9.5 s self-MTP preparation stall, a PLD route that never batched, an MTP cohort locked at width 1, two routes with no stall bound at all, a capsule applied to a row nothing reads), not from the kernels v1 ranked highest. The one kernel win was a bit width v1 could not see.

v1's ten decisions, scored

#	ITEM	DECISION	BUILT	EVIDENCE	VERDICT
1	A1	Route and numerical-law dossier; varlen before speed		M5 GPU · MEASURED	PENDING
2	B1	MoE tiles from the routing distribution		M5 GPU · MEASURED	REJECTED ADOPTED · DEFAULT · B1' 6-BIT
3	A2	Completion-time, geometry-aware prefill bound		M5 GPU · MEASURED	ADOPTED · DEFAULT
4	B2	Exact GVR self-sampling QSA top-k		M5 GPU · MEASURED	OPT-IN

#	ITEM	DECISION	BUILT	EVIDENCE	VERDICT
5	C1	Exact stochastic block verification	●	CPU · REAL CODE	OPT-IN
6	B3	APC value-aware retention	●	CPU · REAL CODE	OPT-IN
7	A3	Capsule representation and metric	●	CPU · REAL CODE	FIXED
8	C2	Routed A8 Gate+Up	○	M5 GPU · MEASURED	DEFERRED
9	C3	Conditioned GDN block algebra	○	M5 GPU · MEASURED	REJECTED
10	D1	Output-aware KV and directional recurrent precision	○	REPOSITORY-RECORDED	PENDING

● built ○ partly ○ not built (measured or deferred instead). Each row links to its status block. Adopted means selected by default in source; none of these is requalified yet.

What v1 got right

- **The stall estimator** stays optimistic after a slowdown: confirmed, fixed by default, and the GPU A/B on identical code shows the gain (A2). [\[S01\]](#) [\[E16\]](#)
- **The capsule metric** reported 0 error for a 55.47% waypoint error: confirmed to the digit and fixed (A3). [\[S08\]](#)
- **Varlen flags are off** in today's Qwen3.8 default: confirmed in source and history. [\[C02\]](#) [\[E10\]](#)
- **Block verification** is a distinct, exact acceptance law with no gain for greedy rows: proven by exhaustive oracle and built opt-in (C1). [\[S04\]](#)

What v1 got wrong or could not see

- **Prize sizes.** The distribution-aware MoE planner recovers about 0.5% of a 35B prefill and GDN block algebra about 1% of a 27B prefill; both rejected (B1, C3). [\[E05\]](#) [\[E07\]](#)
- **The real MoE lever** was a bit width: 6-bit experts in half the layers of the served 35B ran on stock gathers. Built, exact, and now default: 8K TTFT −7% (B1'). [\[E05\]](#) [\[E18\]](#)
- **Serving seams dominated.** At the baseline, Flash-Next's served route let a decoding neighbour wait 6.9-9.5 s while a 16K prompt prefilled; prompt lookup ran B4 as serial B1 rounds; native MTP at B4 lost to ordinary decode. On the final tree the stall is 0.44-0.47 s and MTP B4 is 1.5× ordinary; PLD B4 reached parity with ordinary (55–63 tok/s). [\[E01\]](#) [\[E02\]](#) [\[E15\]](#) [\[E17\]](#)
- **Heat changes the answer.** A route gap seen on the first, cool run was retracted; in two of three arms a later rep drew 34-51% less power. Equal-heat A/B is now a method requirement, and the GPU budget, not temperature, is the limit (chapter 10). [\[E03\]](#) [\[E09\]](#) [\[E13\]](#)

v1's three findings, with GPU data

V1 FINDING	WHAT V2 FOUND	NOW
Source contradicts the varlen story: the Qwen3.8 default disables both varlen flags.	Confirmed. "Varlen" named three mechanisms, and the 1.057x record ran a TensorFold law later found defective. At that varlen-off default the DFlash2 tree is the fastest B1 route (final: 112 vs ordinary 32.9 tok/s); the dossier is still unwritten. [E01] [E15] [E10]	PENDING
The fairness cap stays at 4,096 tokens after a 16x slowdown (synthetic probe).	Confirmed with three repros, plus two more defects (row-blind, pooled kinds). Fixed by default; on the GPU the cost model lowers neighbour p95 by 15%-23% on identical code. The largest stall came from self-MTP preparation the estimator never saw; now bounded. [S01] [E16] [E17]	ADOPTED · DEFAULT
The capsule compiler reports 0 error for a 55.47% waypoint error.	Confirmed to the digit and fixed. A larger serving bug sat next to it: the served capsule was applied to a row nothing reads. [S08] [S09]	FIXED
Several peer ideas are already incorporated.	Confirmed by the sweep's peer survey: most of what peers shipped since 22 September MLX2 already had; three peer bugs applied and were fixed. [E10]	FIXED

WHAT IS STILL OPEN

The GPU numbers are A/B evidence (two or three reps, rotated, one machine), not qualification and not controlled performance runs. Changed defaults (cost-model estimator, 6-bit NAX admission, stall-bounded external routes) need requalification. Four deciding runs have not happened: block verification at temperature 0.7, GVR on a loaded model, value retention on an agent trace, and the PLD B4 plain-round fix. Chapter 11 lists them with the integration findings and open limits.

CHAPTER 2

The M5 Max baseline, and after the sweep

v1 had no Apple GPU. These are the first like-for-like route comparisons through the real HTTP serving path: at the defaults the sweep started from, and again on the final integration tree.

Each campaign starts one server per arm, drives it over HTTP with `scripts/profile_features.py`, rotates arm order across reps, and records TTFT, decode rate, receipts, counter deltas, peak Metal bytes, swap and (where available) IOReport energy. Swapouts did not change in any campaign. Arms differ only by route flags or policy. Three trees were measured: the **baseline** (origin/main 187c6aa23), an **intermediate** integration tree used only for A/Bs whose arms share identical code, and the **final** tree (main) with every fix. Before/after comparisons below are baseline vs final. Speculative outputs differ textually from ordinary decode; the profiler records identity, and the repository classifies such differences as near-ties only under a strict rule (chapter 9). [\[E01\]](#) [\[E15\]](#) [\[E02\]](#) [\[E17\]](#) [\[S11\]](#)

M5 GPU · MEASURED

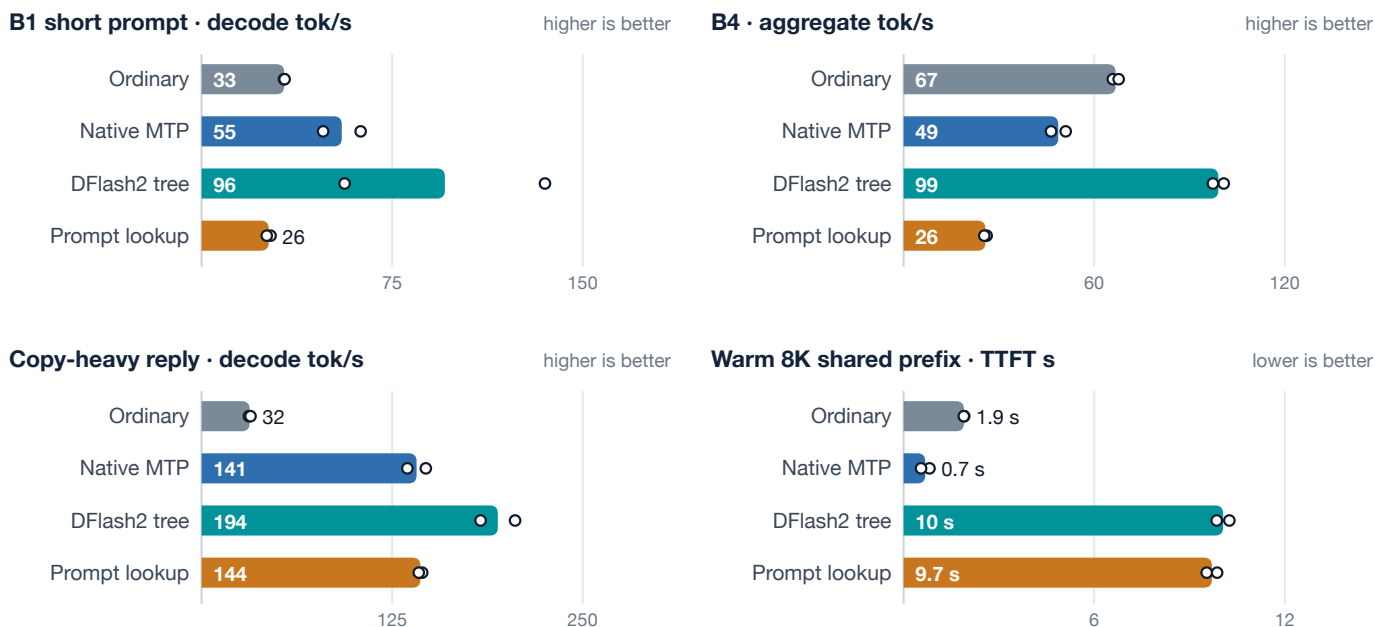


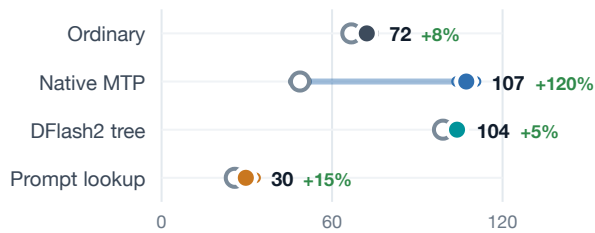
Figure 2.1. The baseline. Qwen3.8-27B oQ4e, max 4 lanes, 256 output tokens. Bars are medians; open dots are the two reps. DFlash2 is the artifact-bound tree default; prompt lookup is the default PLD policy. Source: `q38-routes/summary.json` [\[E01\]](#)

At the baseline DFlash2 won short and copy-heavy decode by the largest margin, and at B4 it was the only route above ordinary (99.1 vs 66.8 tok/s). Native MTP lost at B4 (48.7): a cohort that formed at width 1 (SS-1). Prompt lookup was at or below ordinary everywhere except copy-heavy output, because on a hybrid target it never batched (SPEC-10). On a warm 8K shared prefix MTP was fastest, while DFlash2 and PLD recomputed the whole prompt (section 4.3). [\[S02\]](#) [\[S03\]](#)

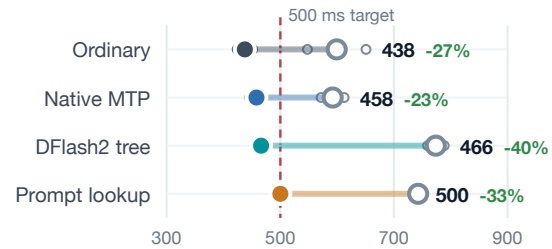
○ baseline 187c6aa23 ● final tree (colour = route or arm)

B4 · aggregate tok/s

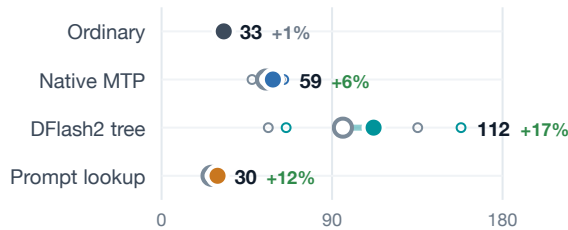
higher is better

**Neighbour gap p95 · ms**

lower is better

**B1 short · decode tok/s**

higher is better

**Warm 8K shared prefix · TTFT s**

lower is better

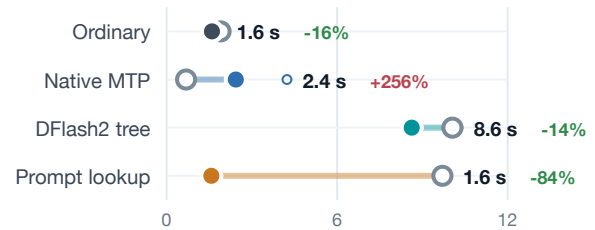


Figure 2.2. The same four routes on the final tree. Hollow grey = baseline median, filled = final median, small rings = individual reps; the percentage is final vs baseline, green when better. Sources: [q38-routes](#), [final-q38-routes](#) [E01] [E15]

What the fixes changed. Native MTP B4 rose to 107 tok/s against ordinary 72.2 once co-arrived prompts formed one cohort (S02, S35). Every route's neighbour p95 is now within about 10 ms of the 500 ms target or under it; DFlash2 and prompt lookup moved most, because S33 put their own prefill loops under the stall bound. Prompt lookup's warm TTFT fell from about 9.7 s to 1.6 s once it recorded recurrent checkpoints. Two things did not improve: PLD's B4 aggregate is still about its own B1 rate (section 4.2), and native MTP's warm hit depth varies between reps (section 4.3), which is why its warm median rose. [S02] [S33] [S35] [E15]

○ baseline ● final tree (colour = route or arm)

Longest pause between a decoding neighbour's tokens (log scale)

lower is better

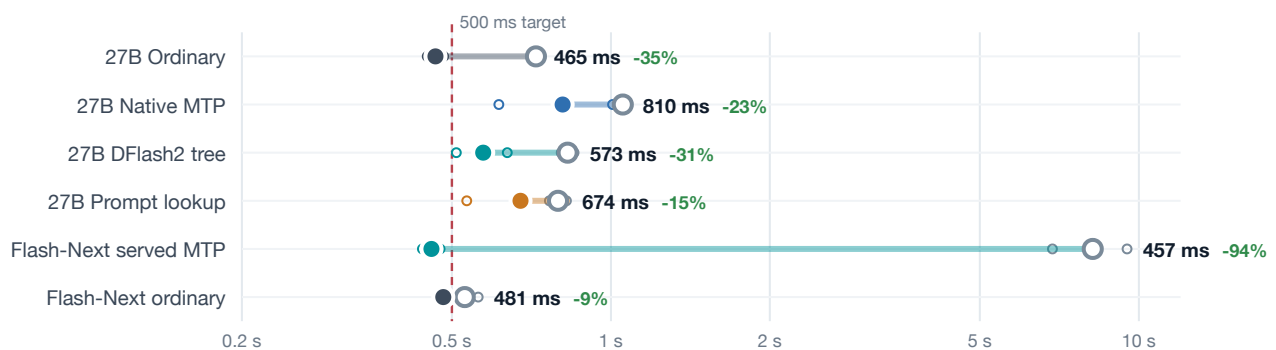


Figure 2.3. The longest pause a decoding request saw while a 12K (27B) or 16K (Flash-Next) prompt was prefilled beside it, per rep. Flash-Next's served native-MTP policy let a neighbour wait 6.9-9.5 s at the baseline; on the final tree it stays under 0.5 s. Native MTP on the 27B still has a ≈ 1 s outlier rep. Sources: [q38-routes](#), [final-q38-routes](#), [fn](#), [final-fn](#) [E01] [E15] [E02] [E17]

Flash-Next and Qwen3.6

FLASH-NEXT UNCENSORED 4-BIT MTP, MAX 8 LANES	SERVED, BASELINE	SERVED, FINAL	ORDINARY, BASELINE	ORDINARY, FINAL
B1 short, decode tok/s	84.2	85.7	60.5	61.3
B8 throughput, tok/s (batch wall time)	92-95	93-101	86-96	88-89
B8 median TTFT, s	6.8-7.0	3.0-3.6	4.2-4.9	5.3-5.8
16K contention: neighbour max gap, s	6.86-9.50	0.44-0.47	0.50-0.56	0.48-0.49
warm 16K TTFT, s	0.69	0.67	2.19	2.20
copy-heavy decode tok/s	229	235	61.2	62.7

The served policy (native MTP, handoff width 3) is about 1.4x ordinary at B1 and 3.7x on copy-heavy output. Its neighbour stall is fixed. At B8 the final tree delivers the same throughput with the median TTFT about halved, because co-arrived prompts are now prepared together instead of in waves. An earlier reading of a B8 "regression" (143-159 → 87 tok/s) was a metric artifact: the aggregate rate was measured from each request's first token, which rewards staggered starts; B8 is measured here as batch wall time, submission to last token. QSA direct4 (baseline only) was neutral against the served radix selector. Qwen3.6-35B-A3B (artifact without an MTP head, ordinary only) at the baseline: B1 136 tok/s, B8 aggregate 251 tok/s, 16K TTFT 4.3 s, neighbour max gap 196 ms, about 0.26 J per token at B1 and 0.17 at B8; its 6-bit NAX result is in chapter 5. [\[E02\]](#) [\[E17\]](#) [\[E04\]](#) [\[E09\]](#)

DEPLOYMENT DRIFT FOUND ON THE WAY

The live 27B service ran a compiled-replay policy from an unmerged worktree that main's adapter refuses. Default, deployed and qualified are three separate states and were three different configurations on 6 October. [\[E09\]](#) [\[E10\]](#)

CHAPTER 3

Scheduling and the prefill stall bound

v1 demonstrated that a running maximum stays optimistic after a slowdown. The sweep confirmed it, found two more defects in the same estimator, found that the worst stall came from paths the estimator never saw, and then measured the fix on identical code.

A2 Completion-time, geometry-aware prefill bound

M5 GPU · MEASURED

ADOPTED · DEFAULT

v1 decision 3 · v1 proposed: Replace the running-maximum stall bound with a conservative completion-time predictor indexed by geometry and state.

BUILT?

Yes: S01 (d28a774aa). PrefillCostModel $t = a + b \cdot R$ per (kind, KV-depth bucket), q90 slope, regime shifts, 2x trust region, 10% margin; MTP preparation bounded and observed. Default estimator cost_model; running_max kept for A/B. Then extended to the external-draft and prompt-lookup prefill loops (S33) and to bursts of self-MTP preparations (S35).

TRIED?

GPU A/B of the two estimators on identical code (E16); CPU closed-loop traces of the real policy (E11); before/final route reruns (E01 → E15, E02 → E17).

MEASURED

Cost model vs running maximum, 12K prompt beside a decoding lane, paired by rep: neighbour gap p95 −15% on native MTP and −23% on ordinary, max −13% and −18%; long-prompt TTFT +6% and +1%; B4 aggregate unchanged. On the final tree every 27B route's p95 is 424-508 ms, and Flash-Next's served stall fell from 6.9-9.5 s to 0.44-0.47 s.

V1 GOT RIGHT

The running maximum stays optimistic after a slowdown (confirmed as SS-2 with three repros).

V1 MISSED

It is also row-blind (a B-row round costs B times the target), pools ordinary/mixed/MTP and shallow/deep samples, and the recorded 500 ms target was not the one enforced. The worst GPU stall was outside the slicer: self-MTP one-call preparation (SS-3), up to 9.9 s on Flash-Next, and two routes had no fairness at all.

NEXT

Requalify routes that slice beside decode (the estimator is now part of route identity). Max-gap outliers remain: native MTP up to ≈ 1 s.

CAMPAIGN ESTIMATOR · LANDED

CAMPAIGN POST-FN · LANDED

CAMPAIGN FINAL-Q38-ROUTES · LANDED

Sources: [S01] [S33] [S35] [E16] [E15] [E17] [E11] [C04]

3.1 What was wrong with the bound

The default-on decode-fairness policy caps each prefill slice so a decoding neighbour waits at most a target (500 ms in serving). Its prediction used the fastest per-row width rate ever observed. The sweep's serving lane reproduced three defects, each with a failing test: **one fast sample pins the cap** after any slowdown; **rows are ignored**, so a B-row round costs about B times the target; and **kinds are pooled**, so ordinary, mixed prefill+decode and self-MTP samples, shallow and 40K-deep, feed one maximum. In the default serving shape the adaptive slicer usually bound first, so the recorded 500 ms target was not the one actually enforced. [E10] [C04]

CPU · REAL CODE

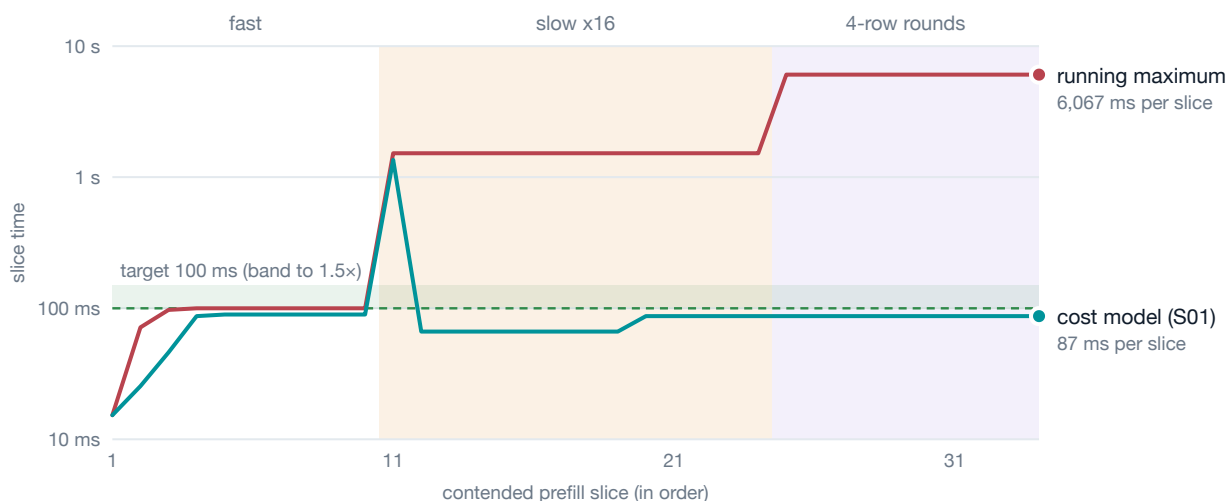


Figure 3.1. The repository's `DecodeTimeFairness` driven in a closed loop on the CPU: a synthetic device with slice time $a + b \cdot \text{rows-width}$ slows 16x after slice 10, then serves four-row rounds. One overshoot slice is unavoidable when the slowdown first appears; after it, the cost model returns to the band while the running maximum keeps the fast-regime width. Synthetic timing law, real policy code. Source: `data/cpu-sims.json` [E11] [S01]

3.2 The fix, and why it does not collapse

S01 replaces the maximum with $t = a + b \cdot R$ per (kind, KV-depth bucket): a is a fixed per-forward cost seeded from recent decode steps, b the 90th-percentile marginal cost per row over a 16-sample ring. A residual above 1.5x twice (or 3x once) drops old samples; four residuals below 0.67 let the bound grow, at most 2x per slice and never beyond twice the largest forward seen since the last shift. The bound is applied after admission, when the real row count and depth are known.

v1's worry was that shorter slices measure a lower rate and cause collapse. With the fixed cost split out, the next width $(T - a)/b$ does not depend on the last width, so the loop converges whenever $a < T$; when $a \geq T$ the slice floors at the grid, debt guarantees progress and `stall_target_unreachable` records it. The price is slower expansion after a speed-up (about 8 slices instead of 2), which the commit states and tests. [S01] [R02] [R03]

3.3 What it does on the GPU M5 GPU · MEASURED



Figure 3.2. The two estimators on the same intermediate integration tree, so only the estimator differs: Qwen3.8-27B, a 12K prompt prefilled beside one decoding lane, two reps per arm (small rings), percentages paired by rep. Source: `estimator/results.json` [E16] [S01]

The cost model lowers the neighbour's p95 gap by 15% on native MTP and 23% on ordinary, and the max gap by 13% and 18%, which brings both p95 values under the target. The cost is paid by the long prompt: its TTFT is

+6% (MTP) and +1% (ordinary), because it gets smaller slices. B4 aggregate throughput is unchanged (MTP 115 vs 115, ordinary 72.6 vs 75.9 tok/s). This is the trade v1 proposed, and it is now measured. [\[E16\]](#)

3.4 The stalls the estimator never saw

On the GPU the worst pause did not come from slice sizing. On Flash-Next's served native-MTP route a residual that fit the uncontended step was prepared in one call beside live decode, and bursts of one-chunk arrivals were prepared back to back with no debt and no rate sample (SS-3): the neighbour max gap was 6.9-9.5 s against 0.53 s on the ordinary route (Figure 2.3). S01 bounds and observes MTP preparation as its own kind; S41 charges each one-call preparation its own fixed cost and depth, and S42 expires stale cost-model samples before a bound is sized. The external-draft and prompt-lookup routes had no fairness at all: their generators run their own prefill loops, built with decode-time fairness disabled, so DFlash2 and PLD neighbours still saw p95 gaps near 700 ms after S01. S33 gives both loops the same bound. [\[E02\]](#) [\[S01\]](#) [\[S41\]](#) [\[S42\]](#) [\[S33\]](#) [\[E14\]](#)

The intermediate Flash-Next rerun then showed the stall reduced but not solved (max gap 3.8-4.1 s), and a B8 run in which co-arrived prompts were admitted in waves (TTFTs of 1.3, 8.9 and 12.2 s). S35 prepares co-arrived self-MTP prompts together after a partial first boundary and bounds the co-arrival hold per sibling; on the final tree the stall is 0.44-0.47 s and the B8 median TTFT about halved (chapter 2). The same hold fixed a related 27B defect: concurrent prompts longer than one prefill chunk had locked the segmented MTP cohort at width 1 (SS-1, S02). [\[E17\]](#) [\[E09\]](#) [\[S35\]](#) [\[S02\]](#) [\[E01\]](#)

CHAPTER 4

Speculation: useful work per verification

Block verification was built and proven exact. The larger GPU losses were elsewhere: a route that never batched, warm prompts recomputed from zero, and, at the very end, a default that stopped the main route from starting.

C1 Exact stochastic block verification

CPU · REAL CODEOPT-IN

v1 decision 5 · v1 proposed: Start with one linear external-draft route, stochastic sampling, a small-vocabulary exhaustive oracle, then GPU.

BUILT?

Yes, opt-in: S04 (exact_verification 'block' on the linear external chain; dense and compact variants; refused with FLY, trees, the continuation pool and adaptive verification). The codex review made acceptance-uniform comparisons strict in block and self-MTP verification (S36).

TRIED?

Exhaustive RNG-branch oracle of the real verifiers (V 3-5, K 2-4, masked, one-hot, p=q, mixed, compact), three wrong variants detected; this review's alignment sweep (E11).

MEASURED

Output law equals the target joint (TV < 1e-9). E[accepted] +5% to +18% on stochastic drafts; identical for point-mass drafts. Gain peaks for weakly aligned drafts: +11.8% at α 0.2, +5.7% at α 0.6.

V1 GOT RIGHT

A genuinely different acceptance law, exact, with no gain for greedy rows; the oracle-first plan was the right order.

V1 MISSED

The default Qwen3.8 route is a tree, where block verification is refused, so the gain is reachable only on the chain route at temperature > 0. An exact self-MTP block verifier already existed but was unreachable and untested (SPEC-05).

NEXT

GPU A/B at temperature 0.7, PLD-free, B1 and B4 chain: accepted tokens per verify and tok/s. Not run.

Sources: [S04] [S36] [E11] [E12] [C06] [R06] [P14]

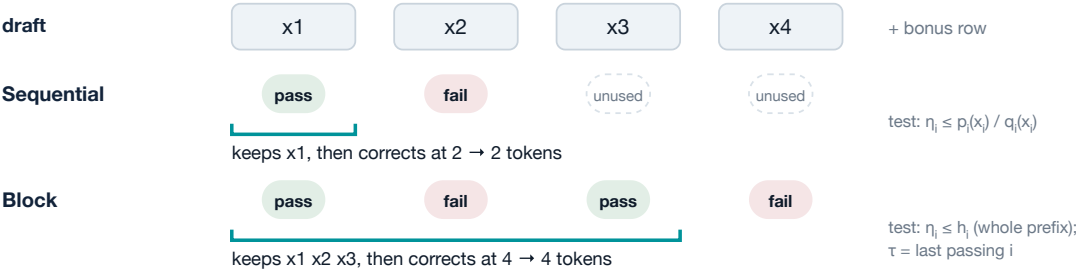


Figure 4.1. One illustrative K = 4 round (Sun et al. 2024, Alg. 2). Under the sequential rule the failure at position 2 ends the round; under the block rule position 3's test uses the whole prefix and passes, so three draft tokens are kept. Decision rule only, not data; with point-mass drafts both rules keep the same prefix.

4.1 How block verification differs

Sequential verification walks the draft and stops at the first rejection. Block verification (Sun et al. 2024, Algorithm 2) computes, for each prefix, the probability mass the target and draft share, and accepts the *last* position whose

test passes; a later pass can rescue an earlier failure, and the correction distribution is scaled to keep the output law exactly the target's. [R06]

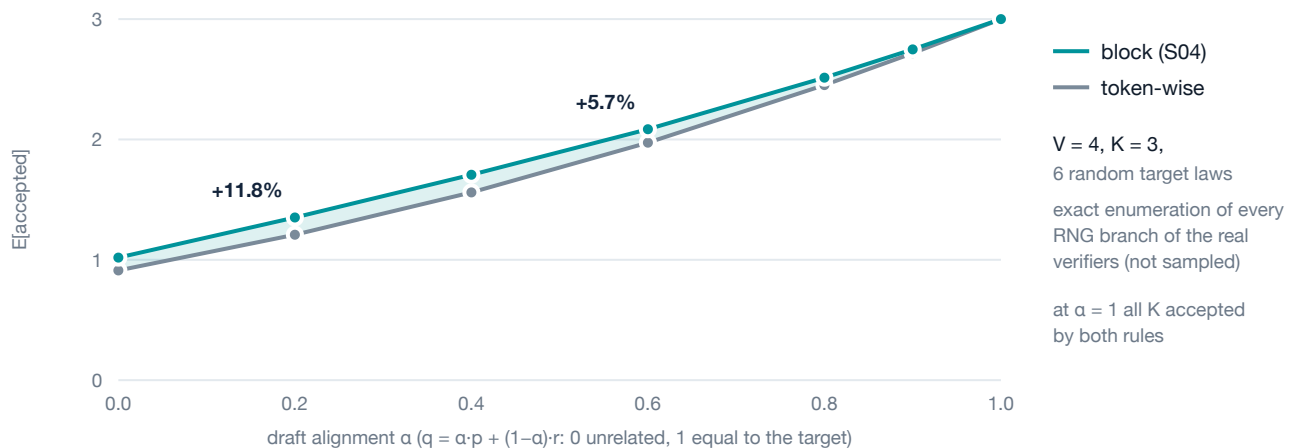


Figure 4.2. Exact $E[\text{accepted}]$ of the repository's `verify_proposals` and `verify_block_proposals`, by exhaustive enumeration of every RNG branch, as the draft law moves from unrelated to identical. The two rules agree at $\alpha = 1$ (all K accepted) and for point-mass drafts; block is never lower. Source: `data/cpu-sims.json` [E11] [S04]

Where the gain is reachable. Only stochastic drafts gain: greedy MTP rows, prompt-lookup rows and any drafter at temperature 0 are point masses, for which both rules accept the same prefix. The default Qwen3.8 DFlash2 route verifies a tree, where block verification is refused (trees are already optimal sample-and-match). So the gain applies to the linear chain route with sampling. That GPU A/B has not been run; the codex review's one finding here (acceptance uniforms compared with $\leq$ instead of $<$) is fixed in S36. [S04] [S36] [E12] **PENDING**

4.2 Prompt lookup on a hybrid target

Before S03: hybrid target, four lanes



4 synchronous target forwards per round; latched lanes still run the verify kernel; replay on partial accept

After S03 + S32: one batched forward



still open: proposal-free rounds also take the hybrid verify path, $\approx 2\times$ an ordinary B4 step

B4 aggregate tok/s, baseline $\rightarrow$ final (q38-routes $\rightarrow$ final-q38-routes)

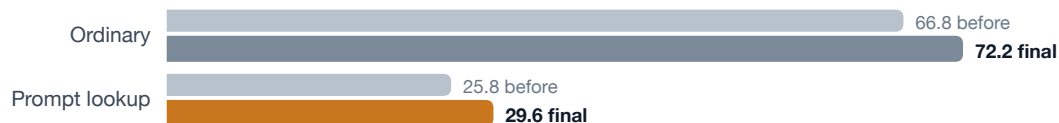


Figure 4.3. Prompt lookup B4 on the hybrid 27B. Batching alone left 814 of 855 cycles as plain-decode rounds at about 114 ms, twice an ordinary B4 step; pipelining those rounds like ordinary decode (82bdcc230) brought the closing run to 55–63 tok/s against ordinary 62–63 (final-pld). Sources: `q38-routes`, `final-q38-routes` summaries, final-pld evidence; cycle counts from the ledger [E01] [E15] [E09]

The PLD batch path accepted only plain and rotating KV planes. Qwen3.8's GDN layers use `ArraysCache`, so every lane ran its own synchronous B1 round; latched lanes with nothing to propose still took the fused GDN verify kernel; every round captured a recovery snapshot; every partial acceptance paid a second forward to replay. S03 routes hybrid lanes through the external route's `HybridVerifyRows` transaction: one target forward per round for the cohort, recurrent rows trimmed instead of replayed. The intermediate GPU run still showed B4 at 27.6–30.4

tok/s with the GPU mostly idle: a lane with undelivered tokens sat rounds out, so the cohort fragmented into one- and two-row forwards. S32 hands out every verified token per poll and keeps the cohort whole. [S03] [S32] [E14] [E10]

This is open: the fix (plain rounds on the plain decode path) was in progress when this review was built. The codex review also found that a failed hybrid round could leave recurrent rows advanced; S37 rewinds them on abort and rebuilds PLD lanes after a failed round. [S37] [E09] **PENDING**

4.3 Warm reuse: why speculative routes recomputed an 8K prefix

The warm workload sends a second request that shares an 8K document prefix with the first. At the baseline ordinary decode restored 6,144 tokens and native MTP 7,168; DFlash2 and prompt lookup restored nothing and paid the full prefill again. [E01] [E15] **M5 GPU · MEASURED**

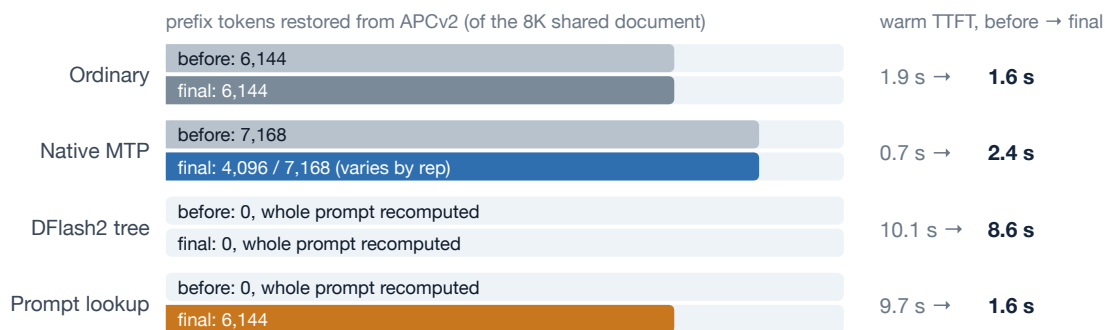


Figure 4.4. Restored prefix tokens per request and warm TTFT medians, baseline and final tree. Prompt lookup now restores 6,144 tokens like ordinary; native MTP restored 4,096 or 7,168 depending on the rep; external draft resumes only from a paired target + draft state, published once at the prompt boundary. Sources: `q38--routes`, `final-q38--routes` results [E01] [E15] [S22]

Prompt lookup lacked recurrent state checkpoints at prefill-chunk boundaries and now records them like ordinary. Native MTP's hit depth is unstable: the same workload restored 4,096 tokens on one rep (TTFT about 4.2 s) and 7,168 on the other (0.6 s); the same instability appeared on Flash-Next's ordinary route (8,192 of 16K once). External draft is limited by design: its generator resumes only from a target-plus-draft state pair, and the only pair it publishes is at the prompt boundary, so a trimmed target-only hit cannot be paired with draft context. Serving reports such a hit as a miss rather than as reuse. Lifting the limit means publishing paired checkpoints inside prefill at a cost of one target KV copy plus one draft-state copy each; this is the warm-reuse ceiling for agent workloads on the DFlash2 default. [S22] [S21] [E15] [E17] **PROPOSED**

4.4 Smaller fixes with measured host cost

- **Discarded tree lattices.** The default tree route prelaunched a draft lattice per lane per round at B2-B4 that the packed cohort never adopted, and never counted the discard (SPEC-01). Prelaunch now runs only for singleton cohorts. [S16]
- **Greedy chain verification** built dense float64 laws over a 248K vocabulary: about 3.6 ms of host work per lane-round at K = 7, about 30 ms per round at B8. A one-hot target accepts a draft iff it is the argmax, so S17 decides in O(K) with the same RNG schedule. The codex review then found that this shortcut skipped the law checks the sampled verifier makes; S40 runs them first. [S17] [S40]
- **Zero residual mass** now draws from the target law instead of raising (SPEC-07). [E10]

4.5 The final A/B caught a broken default

Main commit 66b2f9d1e attached proposal composition to every external drafter with draft distributions, including the pinned Qwen3.8 + DFlash2 pair whose default route is the tree15 policy. The runtime refuses composition on a tree ("proposal composition requires chain verification"), so the default DFlash2 route failed at startup on main.

The first final route campaign found it: the DFlash2 server never came up. S29 defaults composition only on chain routes and lets the tree route accept an explicit opt-out. The A/B that followed shows the tree route is again unaffected: default vs `proposal_composition: false` gives B1 64-157 vs 67-159 tok/s, B4 aggregate 91-101 vs 97-105, copy 192-215 vs 196-223. Composition's default now acts only on chain-route drafters (Muse DFlash2, North EAGLE), where it is unmeasured and needs qualification. [\[S29\]](#) [\[E19\]](#) [\[E09\]](#) **M5 GPU · MEASURED**

CHAPTER 5

MoE, bit widths and kernel Amdahl

v1 ranked a distribution-aware tile planner second. Measured on real routing, it is worth half a percent. The same measurement found a lever ten times larger that v1 could not have seen from source, and it is now built, exact and default.

B1 MoE tiles from the routing distribution

M5 GPU · MEASURED

REJECTED

v1 decision 2 · v1 proposed: Choose NAX tile plans from the actual routed-row histogram instead of the mean rows per expert.

BUILT?

No: measured first (S13), on real routing captured from Qwen3.6-35B-A3B on the M5 Max, against a 36-plan grid, and rejected. The same measurement found the 6-bit gap, which was built (B1').

TRIED?

160 captured routing calls; every emitted and every valid plan timed per call; split-half de-biasing (E05).

MEASURED

Routing is skewed (CV median 1.881, busiest expert ~950 rows vs mean 64) but the mean-based planner is within 3-5% of the best plan per call: recoverable ≈ 0.5% of an 8K prefill (8-11 ms of 1,565 ms); a static M5 retune gets 0.56%.

V1 GOT RIGHT

Tile height is chosen from the mean, and real routing is far from uniform.

V1 MISSED

Skew barely costs time: empty 16-row fragments skip their MMAs (KR-07), so padded-row waste overstates cost. The larger lever was an unsupported bit width (B1').

NEXT

Optional static retune (0.56%). The 6-bit result is in B1'. CAMPAIGN FINAL-Q36-6BIT · LANDED

Sources: [E05] [S13] [C08] [R07] [P01]



Figure 5.1. Rows per expert for one real routing call (the call with the median skew of 160 captured), Qwen3.6-35B-A3B, 2,048 tokens × top-8 = 16,384 routed rows. The planner sees only the mean. Occupancy at fixed tile heights is shown for intuition; the kernel skips empty 16-row fragments, so occupancy is not time. Source: `capture/routing.npz` via `data/cpu-sims.json` [E05] [E11]

Why skew does not become time. Inside a NAX tile, row SIMD groups with no rows skip their MMAs, so wasted multiply work is bounded at 16-row granularity whatever the tile height. What the height trades is the fixed per-tile cost of staging a weight slab and residency. On real routing with coefficient of variation near 1.9, the planner's

choice was within 3-5% of the per-call best of all 36 valid plans, and the de-biased recoverable total was about 0.5% of an 8K prefill: below the 3% bar the research set in advance. [E05] [E10] M5 GPU · MEASURED

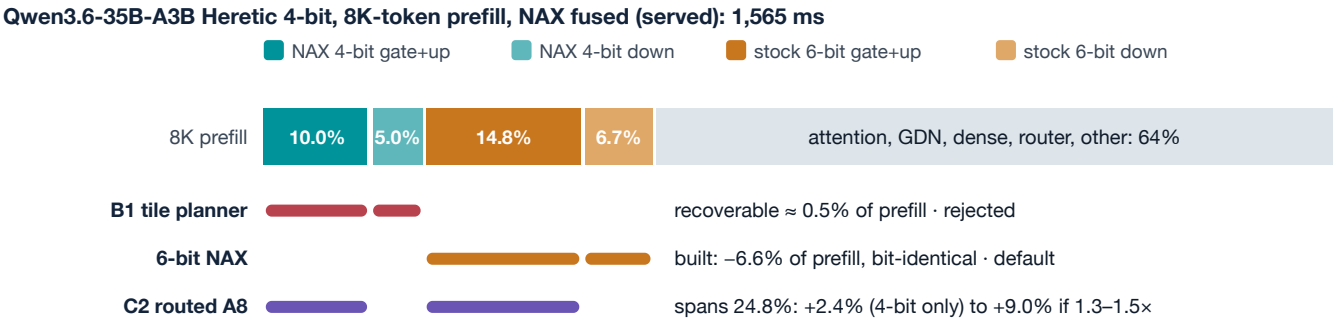


Figure 5.2. The 8K prefill of the served 35B (median of five, NAX fused, before 6-bit support) split by routed expert work. Underlines mark which slices of time each candidate can touch: the histogram planner (B1), 6-bit NAX support, and routed A8 Gate+Up (C2). Shares are sync-corrected per-call kernel times; the NAX on/off end-to-end difference (120 ms) cross-checks them within about 10%. Sources: `analysis-tiles-r1.json`, `nax-6bit/README.md` [E05] [E06] [E20]

5.1 The 6-bit gap, closed

The served Heretic 35B artifact is mixed precision: layers 0-4, 7, 10, ..., 31 and 34-39 carry 6-bit affine experts, and the NAX gather admitted only 4- and 8-bit. Those 20 layers fell back to stock sorted `gather_qmm`: 336 ms, 21.5% of prefill. The estimate was about 84 ms (5.3%) if NAX's 4-bit speedups carried over. [E05]

B1' 6-bit NAX sorted gather (found by B1)

M5 GPU · MEASUREDADOPTED · DEFAULT

v1 proposed: Not a v1 item. B1's measurement showed that 20 of the 40 MoE layers of the served 35B carry 6-bit experts the NAX gather declined: 336 ms, 21.5% of an 8K prefill on stock gathers.

BUILT?	Yes: S25 (46656a8ec) unpacks MLX's affine 2/3/5/6-bit streams in the NAX tile loader; S26 (7e28b1ace) admits 4/6/8 by default, 2/3/5 implemented but refused. Every instantiation runs a bitwise canary against stock <code>gather_qmm</code> and falls back on failure.
TRIED?	Canaries (102 6-bit instantiations, 8/8 real shapes bitwise, a falsifier that drops half the unpack is refused), per-call microbenchmark, an 8K ABBA A/B through the adapter (E20), then the served HTTP A/B on the final tree (E18).
MEASURED	Served 35B, paired by rep: 8K TTFT -7%, 16K TTFT -8%, B8 throughput by batch wall time +4.5%; decode unchanged; logits and 24 greedy tokens bit-identical. Per call vs stock (T 2048): gate+up 1.56x, down 1.29x.
V1 GOT RIGHT	That the 35B MoE kernels left prefill time on the table.
V1 MISSED	v1 looked at tile shape; the time was in a bit width the kernel refused. An earlier 'zero fallbacks' record had been taken on an all-4-bit artifact.
NEXT	Requalify the 35B route and any route serving 6-bit experts (outputs are bit-identical by construction; engagement and receipts change). <code>MLX2_MOE_NAX_GATHER_BITS=4,8</code> restores the old admission.

Sources: [S25] [S26] [E18] [E20] [E05] [C08]

How it stays exact. MLX packs affine 6-bit values as 16 values in 3 words, so some values straddle two words. The tile loader now reads the packed bit stream for widths that are not a power of two, taking a straddling value's high bits from the next word; dequantization is unchanged ($\text{float}(q) \cdot s + b$, rounded once), so each row's arithmetic is the stock gather's. Every kernel instantiation runs a one-time bitwise canary against stock `gather_qmm` before use and returns `None` on failure, so the stock path runs; a falsifier build that drops half the unpack is refused on every 6-bit instantiation. [S25] [E20]

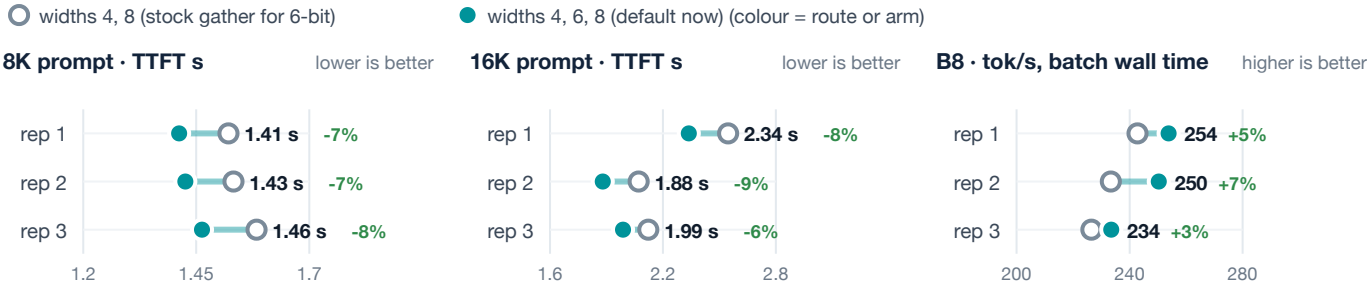


Figure 5.3. Qwen3.6-35B-A3B Heretic through the HTTP serving path on the final tree, three rotated reps; each row pairs the two arms of one rep. Decode is unchanged (B1 155 vs 154 tok/s) and outputs are bit-identical; the B8 gain on the aggregate metric is +4%. Source: `final-q36-6bit/results.json` [E18] [S26]

Per call on captured 35B routing (sums over calls, 7 shuffled interleaved reps): 6-bit gate+up 29.5 → 18.9 ms (1.56x) at T 2,048 and 29.6 → 21.5 ms (1.37x) at T 8,192, down 13.1 → 10.2 ms (1.29x) and 14.0 → 10.8 ms (1.29x); 4-bit gate+up for reference 26.4 → 17.4 ms (1.51x). 6-bit unpacking does not eat the gain. The adapter-level 8K ABBA A/B measured 1,513 → 1,414 ms (1.071x) with every pair faster. Sources: `nax-6bit/microbench.json`, `e2e-ab.json` [E20]

C2 Routed A8 Gate+Up

M5 GPU · MEASURED

DEFERRED

v1 decision 8 · v1 proposed: A narrowly scoped approximate candidate joining dense A8 prefill with routed expert Gate+Up.

BUILT?

No. Amdahl bound from the measured B1 fractions (E06).

TRIED?

Arithmetic on E05's kernel times; no kernel built. Separately, the dense int8 prefill arm on the 27B was refused by that adapter, which declares no int8 prefill scope (fail-closed, correct).

MEASURED

Routed gate+up is 24.8% of the 35B 8K prefill. At 1.3-1.5x: +6.1% to +9.0% if A8 covers every layer (needs a 6-bit A8 path), +2.4% to +3.5% on the NAX 4-bit layers only.

V1 GOT RIGHT

Scope (affine Q4/GS64, plain SwiGLU, Down untouched) and the warning that it is approximate.

V1 MISSED

On this artifact about 60% of gate+up time was in 6-bit layers NAX did not serve; 6-bit NAX (exact) had to come first, and now has (B1'), which also shrinks the time A8 could act on.

NEXT

Not scheduled; re-derive the bound on the post-B1' profile first. W8A8 does not apply to the 27B.

Sources: [E06] [E05] [E09] [C09] [P04]

7 October 2026 | v2 final: source audit + GPU campaigns + sweep

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CHAPTER 6

Sparse selection and GDN prefill

GVR was built, merged opt-in and is exact on the Metal GPU; whether it pays end to end is still open. The GDN block algebra was measured against the chunked scan the adapter already has, and the whole prize is about one percent.

B2 Exact GVR self-sampling QSA top-k

M5 GPU · MEASURED

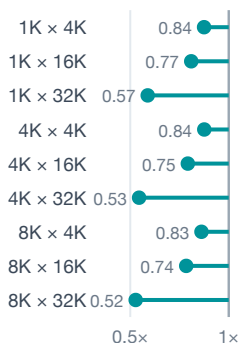
OPT-IN

v1 decision 4 · v1 proposed: Prototype GVR V2's current-row sampling with exact refinement, preserving the block-ID tie law, with a radix fallback.

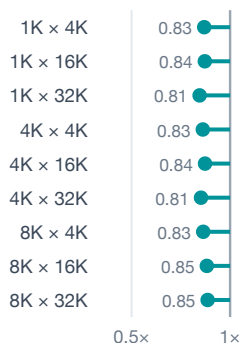
BUILT?	Yes: S12 (41b87aa11 + 0a1c483ca), merged into the integration tree (S31), default off, route labelled unqualified. One Metal kernel: strided samples, four quantile thresholds, canonical tie fill, in-kernel direct8 fallback.
TRIED?	Metal parity 69/69 (kernel = direct8 = law; diagnostics = NumPy mirror). The full synthetic grid: 9 selector, 9 stage-one and 9 adversarial cells, 9 repeats, interleaved arms, no weights (E08).
MEASURED	Bit-exact in every cell. Median over the grid: the selector takes 0.75× direct8's time and the complete stage one 0.83×; at 8K rows × 32K blocks stage one is 59.0 ms vs 69.1 (direct8) and 77.7 (served radix). Heavy ties fall back and are slower than both direct selectors (16-way ties 1.15×, all equal 1.61×).
V1 GOT RIGHT	A distinct algorithm, and the full tie law can be kept exactly.
V1 MISSED	Synthetic wins have not survived loaded models before: direct8 measured +2.7% synthetic but -2.93% loaded, direct4 +0.86%; in E02 the direct4 arm was neutral. Stage one's share of Flash-Next prefill is unmeasured, so the Amdahl ceiling is unknown.
NEXT	Loaded Flash-Next A/B at 32K-128K with P99 selection time; select by default only on a whole-request win. Not run.

Sources: [S12] [S31] [E08] [E02] [C10] [R08] [P02] [P03]

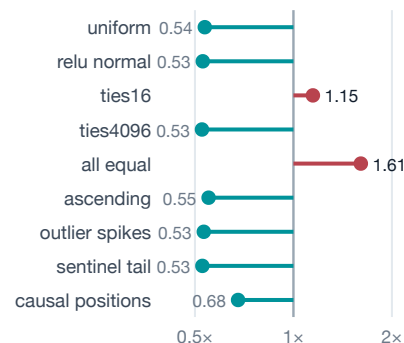
Selector only



Complete stage one



Selector, adversarial



GVR p50 time ÷ direct8 p50 time, log scale; < 1 = GVR faster. Rows × blocks for the grid cells (top-k 512); adversarial cells at 4K rows × 32K blocks. Every cell bit-exact across arms. Synthetic inputs, no model weights.

Figure 6.1. GVR's p50 time as a fraction of direct8's on the M5 Max: nine selector cells, the nine complete stage-one cells (score, select, exact-band refine) and nine adversarial score distributions. Selection is a minority of stage one, so a 0.5-0.85x selector becomes about 0.83x for the stage. Heavy ties defeat the sampled thresholds: 49% of rows fall back with 16-way ties and all rows when every score is equal, and those rows pay for both selectors. Source: [qsa-stage1-gvr-20261006/bench.json](#) [E08] [S12]

How the kernel keeps the tie law. Per row it sorts a strided sample, counts four sample-quantile thresholds in one coalesced pass, gathers the tightest threshold that fits in ascending block order, finds the k-th key among at most CAP candidates, and fills ties from the largest block IDs, so output is canonical without a final sort. Rows where no threshold fits run the direct8 radix inside the same kernel and are flagged. The decision rule is v1's: select it by default only if a loaded-model request gets faster with P99 selection time reported. [S12] [S31] [R08] [C10]

PENDING

C3

Conditioned GDN block algebra

M5 GPU · MEASURED

REJECTED

v1 decision

9 · v1 proposed: Move decay-factor work into diagonal scaling and a block-local triangular solve; guard against FP32 overflow.

BUILT?

No new kernel. Measured the chunked alternative the adapter can already select (gdn_prefill_chunk 8/16) at real 27B shapes (E07), then end to end.

TRIED?

Sequential packed (served) vs chunk 8 vs chunk 16, B 1 and 4, T 512-8192, real A_log/dt_bias; then chunk 16 on the served 27B with --qualification-mode (E09).

MEASURED

chunk16 is 1.19-1.46x per layer call but inexact (state error 3.7-4.5e-3, 0.8-1.6% proxy argmax flips); chunk8 near-exact but slower at T ≥ 2048. The recurrence is ≈ 2.6% of 27B prefill, so the prize is ≈ 1%. End to end, 8K TTFT 8.4-9.0 s vs served 8.3-10.8 s: within thermal noise.

V1 GOT RIGHT

GDN prefill is already chunked; the open question was a better block formulation.

V1 MISSED

The prize: about 1% of prefill. The overflow concern did not appear at real gate values (all errors finite and bounded); the sweep coordinator reports the shipped chunked scan already forms decays relative to the chunk start.

NEXT

None. chunk16 stays rejected (inexact for ≤ 1%).

Sources: [E07] [E09] [C11] [R10]

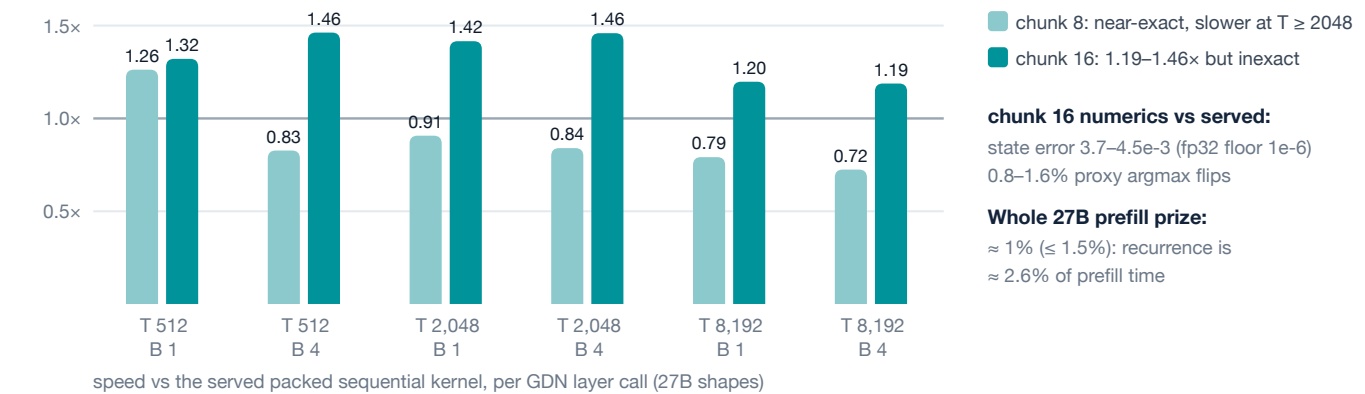


Figure 6.2. One GDN layer's prefill recurrence at Qwen3.8-27B shapes (Hk 16, Hv 48, D 128), real A_log and dt_bias from layer 0. Chunk 16 uses NAX tensor-op inner products; its state error is about 1000x the fp32 reorder floor. Source: c3-gdn-prefill/gdn.json [E07]

The recurrence is about 110 ms of a roughly 2.7 s 2,048-row chunk at B1 (48 GDN layers), so even a free recurrence would save under 3%, and chunk 16 saves about 1%; end to end it was inside the thermal noise (status block above). [E07] [E09] M5 GPU · MEASURED

CHAPTER 7

APCv2, retention and memory

Value-aware retention is built and opt-in. Before it could work, the sweep had to fix the two quantities its value function divides.

B3 APC value-aware retention

CPU · REAL CODE

OPT-IN

v1 decision 6 · v1 proposed: Retain APC entries by expected saved prefill per unique byte rather than category then recency.

BUILT?	Yes, opt-in: S05 (apc_retention_policy 'value'). Its prerequisites came first: per-request hit accounting (S06, APC-2) and one charge per shared checkpoint (S14, COW-1). The codex review then made refused-hit withdrawal and the unleased-bytes estimate exact (S39), and S34 stopped the COW-1 ledger from forcing a device sync on every store.
TRIED?	CPU replay at equal byte budgets (synthetic sessions); GPU evidence that interior checkpoints carry the warm win (E03).
MEASURED	Replay: avoided prefill tokens 23.1K / 36.6K / 76.2K vs 3.0K / 1.8K / 3.3K for the default order at 3K / 6K / 12K-token budgets. GPU: interior checkpoints auto cut warm 8K TTFT to 0.72 s (7,168 cached) from 1.84 s without (6,144).
V1 GOT RIGHT	Categorical rank plus LRU evicts the shared preambles later requests resume from; Marconi's per-byte value is the right shape.
V1 MISSED	The numerator and denominator were both wrong: hits counted lookups, not requests, and kept credit for discarded hits (APC-2); shared checkpoint bytes were charged per entry, up to ~5x (COW-1). APC-1 also hid usable MTP sidecars (Xing: 394/400 misses).
NEXT	Served A/B on a multi-session agent trace; metric: saved prefill ms per resident GiB-s. Not run.

Sources: [S05] [S06] [S14] [S34] [S39] [E03] [E10] [C13] [R14]

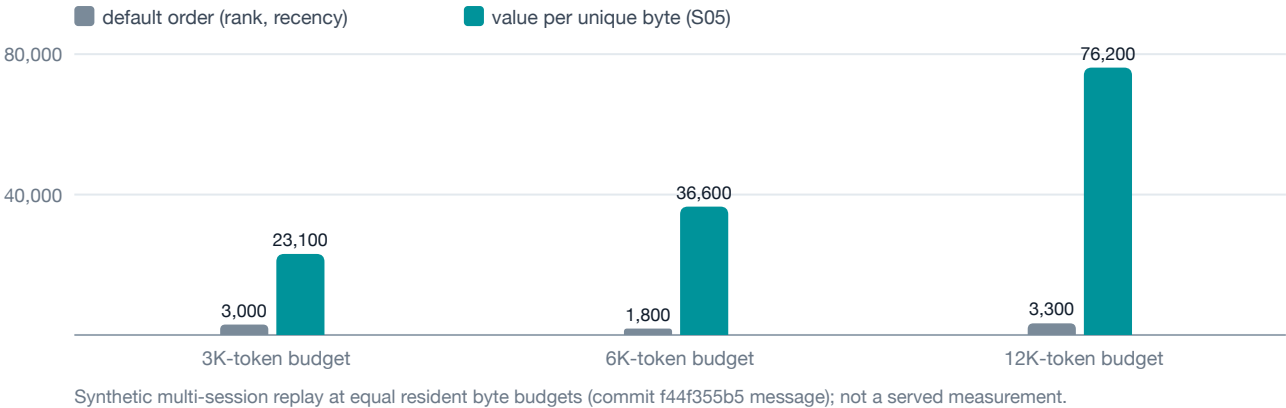


Figure 7.1. Prefill tokens avoided by the cache at three resident budgets. The default order evicts unhit interior checkpoints first and committed prompt boundaries last, so one-off prompt boundaries outlive the shared tool-schema and document checkpoints that later requests resume from. [S05]

7.1 The value function, and what had to be fixed first

```
value(e) =  $\sum_s \text{rate}_s(e) \cdot [C(\text{depth served}) - C(\text{deepest retained ancestor})] / \text{marginal unique bytes}(e)$ 
```

A nested chain is credited only for its increment over the ancestor below it, so a shared prefix is credited once. Two inputs were wrong in source. **Hits** were counted per lookup, not per request: a disk restore counted as miss plus hit, and hits serving threw away kept their credit and even promoted retention rank (APC-2, fixed by S06). **Bytes**: immutable recurrent checkpoint snapshots are aliased into every entry a lane publishes, but were charged once per entry, over-counting the resident budget up to about 5x (COW-1, fixed by S14). [\[S06\]](#) [\[S14\]](#) [\[E10\]](#) [\[R14\]](#)

7.2 Interior checkpoints carry the warm win on the GPU

On the 27B ordinary route, interior checkpoints auto restored 7,168 of 8K shared tokens and cut warm TTFT to 0.72 s from 1.84 s without them (6,144 restored). The same arm was slower cold (not controlled for heat): 10.0 s vs 8.7 s TTFT on an 8K cold prompt. Retention policy decides whether such checkpoints survive; the Hermes incident on 6 October (8 GiB APC evicting interior checkpoints, 22-30K prompts, 50 s TTFT) is the field case. [\[E03\]](#) [\[E10\]](#) **M5 GPU · MEASURED**

7.3 Other state findings

ID	FINDING	STATUS
APC-1	A deeper entry without an MTP sidecar hid a usable shallower sidecar entry, so self-MTP got a full miss (Xing 20x20: 394 of 400 misses were sidecar-missing).	FIXED [S21]
APC-3	Warm continuations cut prefill at different positions from cold, so logprobs differed at near-ties. With the emulated fix they were bit-identical.	FIXED [S20]
COW-2/3	Borrowed-recovery restore accepted a trimmed-then-rewritten cache; approximate-KV staging mutated its source.	FIXED [E10]
PAGED-1..7	Paged index, bridge and private cache (no serving caller yet): failed store bricked the index, dtype unchecked on restore, poisoned writer accepted appends; no eviction and four smaller items.	PAGED-1/3/4 FIXED others open [E10]

CHAPTER 8

Semantic capsules

v1 found a metric that reported a perfect compile for a 55% error. The sweep confirmed it and found that the served capsule had been applied to a row nothing reads.

A3 Capsule representation and metric

CPU · REAL CODE

FIXED

v1 decision 7 · v1 proposed: Make the compiler's reconstruction objective match what it claims before adding capsule capacity.

BUILT?

Yes: S08 (segment sums, waypoint-optimal basis, honest metrics). Then the larger serving fix S09 (CAP-2, CAP-3, CAP-7), S15 (CAP-6) and the bind/mount law (CAP-4/5/8).

TRIED?

v1's own probe as a regression test; 54 compiler cases; a real ServingEngine repro on the tiny Qwen3.8 model.

MEASURED

v1 probe: reported error 0, real waypoint error 55.47%, now exact. CAP-2: the served capsule landed on the second-to-last prompt row; at the shipped layer 31 of 32 the output was bit-identical to ordinary decode while the receipt said observed_used.

V1 GOT RIGHT

The metric mismatch, to the same numbers (55.47%, 33.69°).

V1 MISSED

The serving seam: the M3 mechanism gate injected on the full prompt's last row, serving on a different row. The bus still has no production caller (CAP-9) and no quality claim exists.

NEXT

Re-run the mechanism gate through ServingEngine; persona and routing capsules with the Part B controls.

Sources: [S08] [S09] [S15] [E10] [C14] [C15]

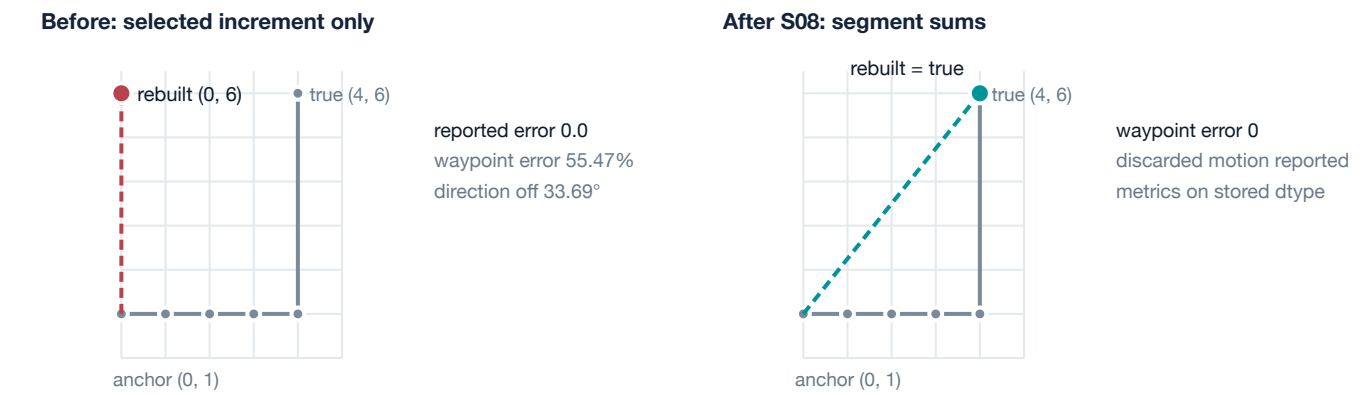


Figure 8.1. v1's probe: states (0,1) ... (4,1), (4,6), keep one transition at rank 1. Before, the stored increment was the single largest step, so the rebuilt waypoint landed at (0,6), a state that never occurred, with a reported error of 0. After S08 each stored increment is the sum of all movement since the previous waypoint and the basis is the SVD of waypoint displacements; the probe is exact. [S08] [C14]

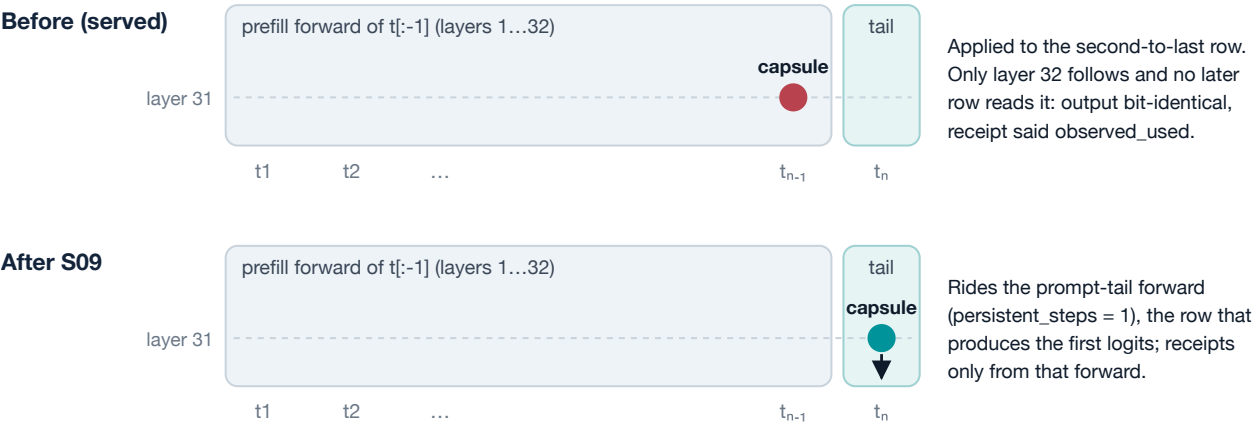


Figure 8.2. The batch generator prefills `tokens[: -1]` and feeds the last prompt token in the first decode step. A final-row intervention passed as a prefill argument therefore amended the second-to-last row. At injection layer 31 of 32 only one more layer reads that row's state and its logits are discarded, so the output was bit-identical to ordinary decode while the receipt claimed `observed_used`. [S09] [E10]

The same seam affected the neural-concept bridge reachable through `--semantic-memory`, and the M3 mechanism gate passed because it injected on the full prompt's last row, a different seam from the one serving executes. S09 moves the memory onto the prompt-tail forward with a one-step lifetime and advances receipts only from a forward that applied it with `gate > 0`. With capsules acting only on the tail, every restorable row is exact ordinary state, so S15 lets capsule requests reuse an ordinary prefix. [S09] [S15] CPU · REAL CODE

8.1 What capsules could be for

The capsule lane ranked ten uses, each with a common control set: ordinary reference, zero gate (bit-identical), random direction at equal norm, wrong-scope capsule, negated or shuffled capsule, and an equal-token text baseline. The top four are prototypes for this week, all offline first: **persona and style capsules** for agents with no prompt tokens; **read-only routing capsules** whose hidden-state prototypes choose a lossless decode policy; **capsule-sourced PLD corpora** for compacted context; and **operator steering capsules**. None has a quality claim yet. [E10] [R17] [R18] PROPOSED

D2

Learned capsule composition / HySparse2

PROPOSED

DEFERRED

v1 proposed: Larger research bets once foundations give trustworthy comparisons.

BUILT?

No. HySparse2 held since 1 October; the capsule bus has no production caller (CAP-9).

TRIED?

Use cases ranked with a common control set (section 8.1).

MEASURED

Nothing measured; the CAP-2/3 lifetime prerequisites are now in place.

V1 GOT RIGHT

Quality, off-context behaviour and composition before capacity.

V1 MISSED

—

NEXT

Persona and routing capsules on the 9B, offline, with the Part B controls.

Sources: [E10] [C17] [R17] [R18] [R20]

CHAPTER 9

Route law, precision and qualification

v1 said yesterday's varlen story contradicted today's defaults. It did, and the record shows why: "varlen" meant three different mechanisms.

A1Route and numerical-law dossier; varlen before speed

M5 GPU · MEASUREDPENDING

v1 decision 1 · v1 proposed: Freeze one authoritative ordinary law and compare every prefill / verify / handoff shape against it before restoring faster packed prefill.

BUILT?

Partly. No single dossier was produced. The sweep made the external varlen pair one operator-owned default group (S19), corrected tree route labels, let the tree route qualify on observed tree rounds, and put external-draft prefill under the stall bound (S33).

TRIED?

Four-route HTTP A/B at the 187c6aa23 defaults (E01), on an intermediate tree (E14) and on the final tree (E15); a composition A/B on the tree route (E19).

MEASURED

Final DFlash2 tree: B1 short 112 tok/s vs ordinary 32.9; B4 aggregate 104 vs 72.2; 8K long-prompt decode 31.0 vs 31.2 (baseline 26.3 vs 31.0); neighbour p95 759-789 → 463-469 ms. A main commit's composition default stopped the default tree route from starting; the final A/B caught it and S29 fixed it.

V1 GOT RIGHT

Varlen flags are off in today's default and the 1.057x history is not today's route: confirmed in source and history (C1, C5 of the history digest).

V1 MISSED

'Varlen' names three mechanisms (native N20 route, external varlen prefill + dense-MLP compaction, ragged-verify layout). The sharper gaps are SPEC-03 (the B5+ leg jumps from 16 to 40-64+ target rows with no evidence) and SPEC-02 (the near-tie contract cannot fail on a small systematic drift).

NEXT

Write the dossier from E01 and E15; add a B5+ cell; make the near-tie gate rate- and shift-aware; requalify the external routes, whose fairness changed (S33).CAMPAIGN POST-Q38-ROUTES · LANDEDCAMPAIGN FINAL-Q38-ROUTES · LANDEDCAMPAIGN DFLASH-COMP · LANDED

Sources: [E01] [E14] [E15] [E19] [S19] [S29] [S33] [C02] [C03]

9.1 Three things called varlen

MECHANISM	EVIDENCE	DEFAULT
Native N20 packed/paged route	Deterministic 10-minute traffic A/B: 7.44% slower overall; only one geometry won (1.045x). Wins scale with removed padding (1.06x at 20%, 1.52x at 40%).REPOSITORY-RECORDED	Off, research route
External varlen prefill + dense-MLP compaction (DFlash2)	1,680 requests, zero mismatches, 1.057x decode, but both arms ran the TensorFold numerical law later found defective.REPOSITORY-RECORDED	Off in the artifact-bound default; one operator-owned group (S19)
RaggedVerifyLayout for self-MTP	Scoped qualification for Qwen3.8 B4 self-MTP physical varlen.REPOSITORY-RECORDED	Contract on, self-MTP only

The instruction "varlen default on" on 5 October referred to the second row; the 6 October artifact-bound default disabled both of its flags and changed pairwise selection from batched to host. Docs said tree verification was

explicit-only while code made it the default; the sweep corrected the labels. [\[E10\]](#) [\[S19\]](#) [\[C02\]](#) [\[C03\]](#)

9.2 The near-tie rule and its blind spot

A width-greater-than-one comparison counts a divergence as a near-tie when the shared prefix, the unordered top two candidates and both margins (at most 0.5 nat) agree and functional oracles pass. Speculative routes in E01 diverge textually from ordinary (DFlash2 B4: 1/8 identical); the rule exists because such ties are routine. The spec lane showed the rule cannot fail on rate or cross-arm shift: a small systematic state perturbation always appears first as a near-tie and passes (SPEC-02). A complementary check should count divergences per thousand tokens and compare margin distributions across arms. [\[E01\]](#) [\[E10\]](#) [CPU · REAL CODE](#)

9.3 False greens in the qualifier

Several default-on mechanisms (Flash-Next fused GDN prefill, MoE weighted sum, lane matmul on covered routes, base decode fairness) were selected but never required, so a route could qualify without them running (CQ-4). S18 requires and observes them as run deltas; older observations that read absolute counters, which a startup probe could satisfy, are now deltas too. [\[S18\]](#)

9.4 Precision objectives

v1's last decision moves from numerical law to precision: choose KV and recurrent-state precision by what the output loses, at equal total bytes. The sweep did not build it; it closed the one production risk v1 attached to it.

D1

Output-aware KV and directional recurrent precision

REPOSITORY-RECORDED

PENDING

v1 decision 10 · v1 proposed: Choose KV and recurrent-state precision by output error at equal total bytes, not tensor MSE.

BUILT?	No.
TRIED?	Sweep audit of the related cache-identity risk v1 raised (first-block channel-norm calibration).
MEASURED	Risk refuted for production: the first-block KVarN calibration is never enabled on a serving path and APCv2 refuses to store approximate caches (latent hygiene only). Recurrent codecs int8-row-v1 and fp16 GDN state remain off.
V1 GOT RIGHT	Output-aware objectives are the right test for any new codec.
V1 MISSED	The cache-identity risk it flagged cannot occur in production today.
NEXT	Not scheduled; behind the open items in chapter 11.

Sources: [\[E10\]](#) [\[C12\]](#) [\[R11\]](#) [\[R12\]](#) [\[R13\]](#)

CHAPTER 10

Heat, energy, the governor and idle

A laptop GPU changes its clock under load, sleeps between requests and draws power someone pays for. None of it is visible in source; all of it changed how the sweep measured, and the power half is now a built, measured control.

10.1 Equal-heat A/B is a method requirement M5 GPU · MEASURED

The first route campaign recorded 8.4 s cold TTFT for ordinary on an 8K prompt and about 10 s for each speculative route, which read as a speculative prefill penalty. It was retracted: ordinary ran first, on a cool die. Run hot, all four routes took about 10 s. In two of the three arms of the feature campaign, the later rep drew 34-51% less power for 9-25% less throughput; the third arm did not change state. Sequential arms therefore confound route with heat. [\[E09\]](#) [\[E03\]](#)

Sequential arms: the first arm runs on a cool die



Retracted claim: 8K cold TTFT ordinary 8.4 s vs ≈ 10 s for the speculative routes looked like a route gap; it was the first, cold run. Hot, all four routes took ≈ 10 s (FINDINGS-GPU).

Strict alternation (thermal_map.py): every arm sees the same thermal history



All arms loaded at once, one per process; cold $\rightarrow$ hot $\rightarrow$ cooldown $\rightarrow$ duty-cycled phases; IOReport joules bracket every segment.

Figure 10.1. Running arms one after another hands the first arm a cool die. `thermal_map.py` loads every arm at once and drives them in strict alternation through cold, hot, cooldown and duty-cycled phases, bracketing every segment with IOReport energy reads. [\[S11\]](#) [\[E09\]](#)

10.2 DVFS trades throughput for energy, unevenly M5 GPU · MEASURED

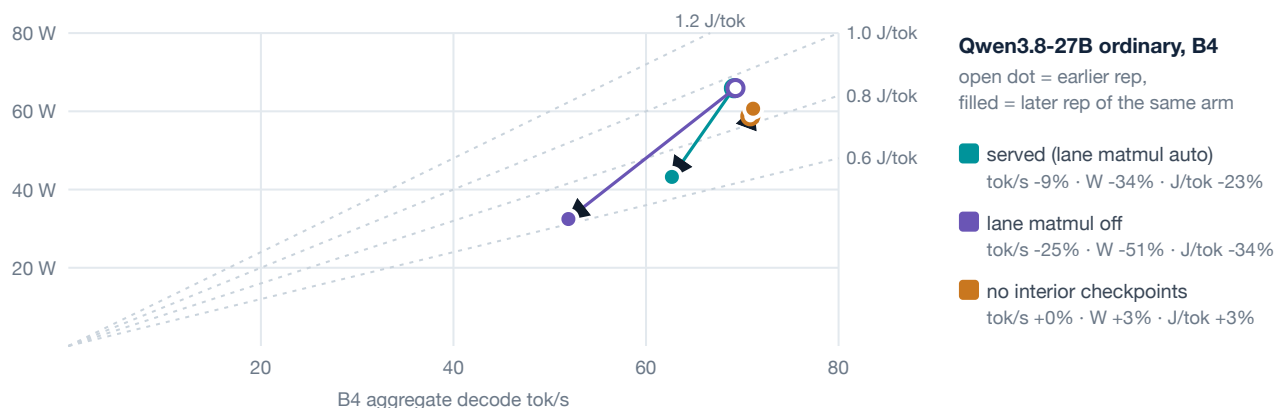


Figure 10.2. B4 decode on the 27B: each arrow joins an arm's earlier and later rep. Dotted rays are constant J per output token (GPU + DRAM). As macOS lowered the mean GPU P-state (10.8 $\rightarrow$ 8.1 served, 11.2 $\rightarrow$ 6.1 lane matmul off), power fell faster than throughput: 23-34% less energy per token. Source: `q38-features/results.json` [\[E03\]](#)

This is the observation behind a governor: on this hardware the fastest operating point is not the most efficient one, and the gap is large. At B1 the 27B drew about 1.1-1.26 J per token; at B4 0.75-1.1; the 35B MoE 0.26 at B1 and 0.17 at B8. [E03] [E04]

10.3 Forty minutes at equal heat M5 GPU · MEASURED

The thermal map drove the three 27B routes in strict alternation from a cool die through 25 minutes of sustained load, 7 minutes idle and 10 minutes duty-cycled. The very first prefill ran at 892 tok/s drawing 89 W of GPU power; after a burst of about 90 s every route settled on a plateau near 654 tok/s at about 40 W, and stayed there for the whole hot phase. The die plateaued at 63-66 °C, so the limit is a power budget, not temperature; after the load stopped it cooled from 65 to 44 °C in about 100 s. The burst is a latency resource: spend it on interactive TTFT.

[E13] [E09] M5 GPU · MEASURED

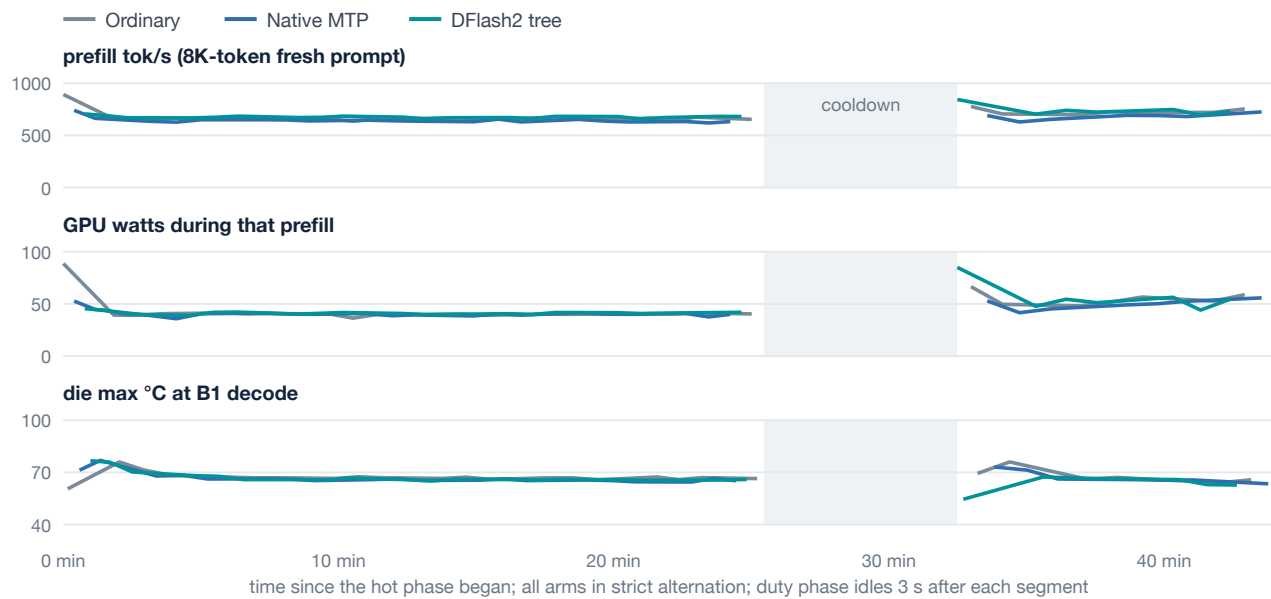


Figure 10.3. Thermal map of Qwen3.8-27B ordinary, native MTP and DFlash2 in strict alternation (complete): each cycle runs, per route, a fresh 8K prefill, a B1 and a B4 decode. The cool-die burst lasts about 90 s; the plateau after it is what longer benchmarks measure. Source: `thermal-q38/segments.jsonl` [E13]

Idle gaps cost energy. With 3 s of idle after each segment, prefill ran +7% to +10% faster than under sustained load but used +20% to +25% more energy per token: the GPU returns to a higher P-state after each gap. For a governor this means that pacing work into bursts is not free, and that the most efficient operating point is steady, moderately clocked load. [E13]

Routes at equal heat. Sustained B1 decode: ordinary 30 tok/s at 1.18 J/token, native MTP 41 at 1.12, DFlash2 55 at 0.82. B4: ordinary 68 at 0.55, MTP 90 at 0.42, DFlash2 84 at 0.46. Speculation is also the energy-efficient choice here: fewer target passes per token. MTP's B4 rate is far above its E01 result because this map's B4 prompts are short and fit one prefill chunk, which the SS-1 cohort lock does not affect. Only native MTP B1 lost rate as the map went on (-14.5% hot vs cold). [E13] [E01]

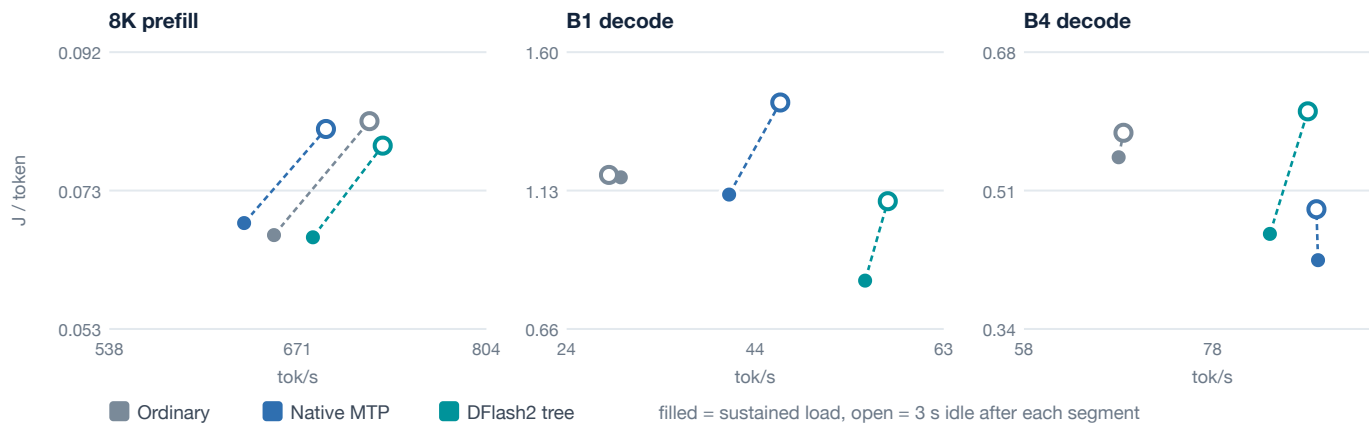


Figure 10.4. Each route's operating point under sustained load (filled) and with 3 s of idle after every segment (open), at equal heat. Idle gaps let the GPU clock back up: a little more speed for markedly more energy per token. Source: [thermal-q38/summary.json](#) [E13]

ROUTE	WORK	COLD TOK/S	HOT TOK/S	DUTY TOK/S	HOT VS COLD	GPU W COLD → HOT	J/TOKEN COLD → HOT	DIE °C COLD → HOT
DFlash2 tree	B1 decode	54.4	54.8	57.1	0.6%	40.6 → 30.8	0.880 → 0.823	76 → 66
DFlash2 tree	B4 decode	82.9	84.5	88.6	1.9%	27.4 → 26.6	0.480 → 0.457	76 → 66
DFlash2 tree	8K prefill	690	682	731	-1.1%	43.8 → 41.7	0.068 → 0.066	73 → 63
Native MTP	B1 decode	47.9	41.0	46.2	-14.5%	40.7 → 34.0	1.130 → 1.117	71 → 65
Native MTP	B4 decode	89.6	89.7	89.5	0.1%	30.3 → 26.5	0.480 → 0.424	75 → 66
Native MTP	8K prefill	664	633	691	-4.6%	44.6 → 40.0	0.072 → 0.068	67 → 63
Ordinary	B1 decode	31.0	29.9	28.7	-3.7%	17.8 → 17.9	1.154 → 1.176	72 → 66
Ordinary	B4 decode	66.7	68.1	68.7	2.2%	26.7 → 26.5	0.566 → 0.551	69 → 65
Ordinary	8K prefill	657	654	721	-0.4%	39.2 → 40.3	0.065 → 0.066	67 → 63

Cold: median of the first three cycles of the sustained phase, which starts on a cooled die; hot: median of its last three; duty: median of the last three cycles run with 3 s idle after each segment. Rates are tokens per second of the segment; J/token is GPU + DRAM. Run from the sweep tree 38129cc92, before the S01-S03 scheduling fixes. Source: [thermal-q38/summary.json](#) [E13]

10.4 Publish power M5 GPU · MEASURED

Pierre's proposal: publish energy and thermal state from the server, and let an operator tune power so that a fleet can reduce or increase consumption to match grid load. The publishing half (S10, default off) is an unprivileged sampler that reads IOReport energy per subsystem, GPU P-state residency and die temperatures once a second,

without touching MLX or Metal, and exports watts, integrated joules, active ratio, mean P-state, die temperature and a rolling J per output token through `/metrics` and `/v1/status.power`, plus a per-request energy estimate. The estimate is an allocation rule (output-token share of each interval's GPU + DRAM joules), not a meter. **It costs nothing measurable:** with telemetry on, B1 decode ran 32.7 vs 32.8 tok/s off, and B4 aggregate 79.8 vs 76.8 (three reps; the B4 difference is noise in telemetry's favour). The codex review hardened the native samplers: NULL results are checked, sampling is serialised, and the attribution window is bounded in seconds (S38). [S10] [S11] [E22] [S38]

10.5 The governor: built, measured, and one diagnosis corrected M5 GPU · MEASURED

The control half is built too (S27, default off). macOS exposes no userspace DVFS or power-limit control, so the governor steers measured GPU + DRAM power by shaping work: `max_throughput` actuates nothing; `efficient` holds the idle admission window so near-simultaneous arrivals start as one wider cohort (B4 is 2.1-2.6x more efficient per token than B1 in the thermal map); `budget` runs a ramped closed loop that rests the scheduler between rounds first and caps lanes last. An operator or a grid signal changes mode and watts at runtime through `POST /v1/admin/power`; every receipt records what the governor did. [S27] [S28] [E13]

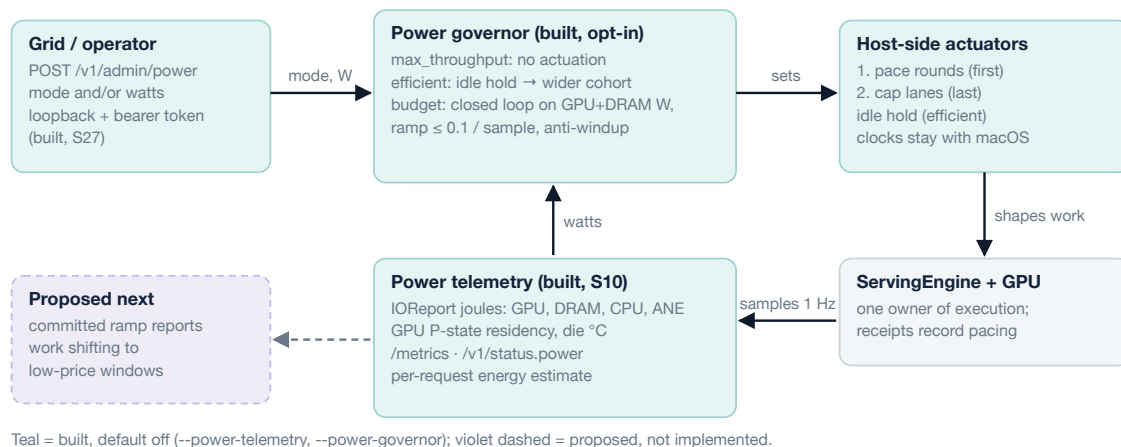
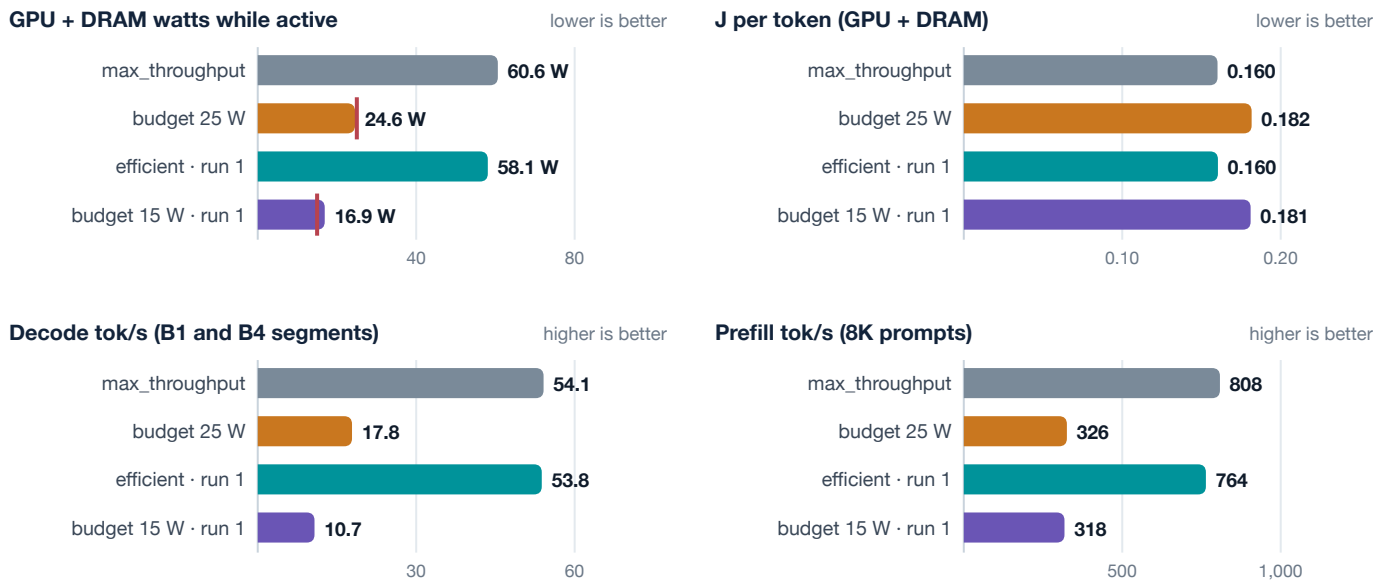


Figure 10.5. What exists and what is proposed. The levers are host-side only (pacing, lane cap, idle hold); clocks stay with macOS. Token outputs change only through batch width, and receipts say so when a request ran under a cap or hold. [S10] [S27]



Red tick = the budget. Time-weighted over every hot segment of strict rotation; 'run 1' rows are from the first governor campaign, before anti-windup.

Figure 10.6. Qwen3.8-27B ordinary, every mode loaded as its own server and driven in strict rotation (prefill, B1 and B4 segments) with power telemetry on. Final-tree rows: max_throughput and budget 25 W after anti-windup (15 min); run-1 rows: efficient and budget 15 W from the first campaign (20 min). Sources: `final-governor`, `governor` segments [E23]

The budget holds. Under a 25 W cap the server averaged 24.6 W against 60.6 W uncapped (−59%); the first run's 15 W cap held 16.9 W. The price is throughput: decode −67%, prefill −60%, and energy per token +13%, because pacing leaves the GPU idle but not free. So `budget` is a reliable hard power cap, a demand-response lever, not an efficiency mode. `efficient` used 4% less energy per token than `max_throughput` on this workload by this document's aggregation (the coordinator's ledger, which counts cycles differently, has about 2%); small, from one 20-minute run, and on a workload whose B4 requests already arrive together. [E23] [E09]

CORRECTION: BUDGET THROTTLING OF B1 IS RIGHT, NOT WINDUP

The first governor run paced B1 decode to about 11 tok/s under a 25 W budget, and the sweep read that as integral windup: the governor reads host-wide power, and an idle budget-mode server could tighten on another server's load. Anti-windup was added (S28: tighten only for intervals in which this engine ran a lane). The final run showed the first reading was mostly wrong: B1 decode draws 19 W of GPU plus 18 W of DRAM, about 37 W in all, because memory-bound decode keeps DRAM busy while the GPU idles at a low P-state. A 25 W GPU + DRAM budget therefore has to throttle it, and does: B1 ran 12.1 tok/s at 15 W against 30.5 tok/s uncapped. Anti-windup stays, because on a shared host the windup path is real. [S28] [E23] [E09]

Extending the idea. Three additions make it useful for grid response rather than only for dashboards. (1) **An operating curve per route:** the thermal map yields tok/s against watts for each route at equal heat; a route switch (for example ordinary to DFlash2 at B1, 1.18 vs 0.82 J/token) is itself a lever. (2) **Committed ramps:** demand-response programs pay for predictable reductions; the controller already ramps at most 0.1 of its throttle per sample and reports the window mean and whether it is within budget, but it does not yet commit to a ramp rate or report achieved vs requested watts per interval. (3) **Work shifting, not only throttling:** deferrable jobs (batch, eval, cache warming) queued for low-price windows, which raises consumption when the grid has surplus. (1) and (3) are proposals. **PROPOSED**

10.6 GPU keep-warm across idle gaps M5 GPU · MEASURED

A peer reported that Apple GPUs drop into an idle power state after about 1.5 s without work, and that the next command buffer stalls for up to about 1.7 s. S07 adds an opt-in ticker that submits a one-element add every 0.5 s, only within N seconds after real work, never while a lane is stepped. Measured on the 27B, the wake penalty on the M5 Max is about 0.3 s after 20 s of idle (TTFT 4.52 → 4.18 s), 0.15 s after 6 s, and nothing after 2 s, far below the peer's figure. The ticker's cost is power (the GPU never reaches its idle state inside the window), so it stays opt-in. [S07] [P24] [S23] [E21]

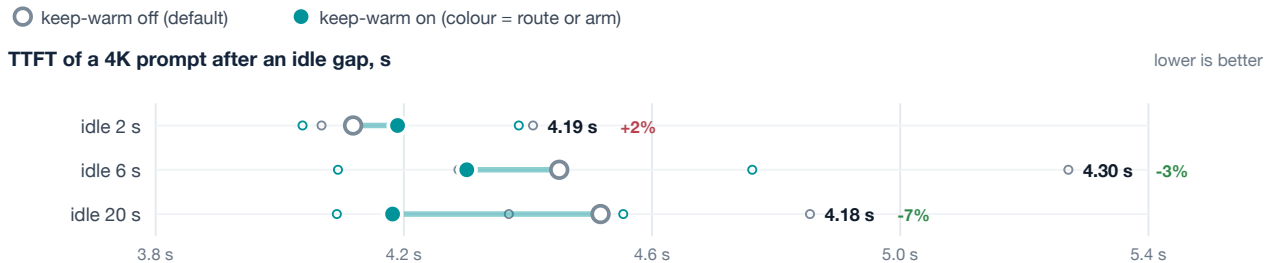


Figure 10.7. A fresh 4K-token prompt after an idle gap, keep-warm off vs on; three reps of three requests per cell (rings are rep medians). Source: `keepwarm/summary.json` [E21] [S07]

CHAPTER 11

Roadmap status, integration findings, plan

Every item in one table, with the final evidence. Then what integrating the sweep broke and fixed, what stays open, and a plan ordered by measured prize.

ITEM	WORK PACKAGE	EVIDENCE	VERDICT	NEXT STEP
A1	Route and numerical-law dossier; varlen before speed	M5 GPU · MEASURED	PENDING	Write the dossier from E01 and E15; add a B5+ cell; make the near-tie gate rate- and shift-aware; requalify the external routes, whose fairness changed (S33).
A2	Completion-time, geometry-aware prefill bound	M5 GPU · MEASURED	ADOPTED · DEFAULT	Requalify routes that slice beside decode (the estimator is now part of route identity). Max-gap outliers remain: native MTP up to ≈ 1 s.
A3	Capsule representation and metric	CPU · REAL CODE	FIXED	Re-run the mechanism gate through ServingEngine; persona and routing capsules with the Part B controls.
B1	MoE tiles from the routing distribution	M5 GPU · MEASURED	REJECTED	Optional static retune (0.56%). The 6-bit result is in B1'.
B1'	6-bit NAX sorted gather (found by B1)	M5 GPU · MEASURED	ADOPTED · DEFAULT	Requalify the 35B route and any route serving 6-bit experts (outputs are bit-identical by construction; engagement and receipts change). MLX2_MOE_NAX_GATHER_BITS=4,8 restores the old admission.
B2	Exact GVR self-sampling QSA top-k	M5 GPU · MEASURED	OPT-IN	Loaded Flash-Next A/B at 32K-128K with P99 selection time; select by default only on a whole-request win. Not run.
B3	APC value-aware retention	CPU · REAL CODE	OPT-IN	Served A/B on a multi-session agent trace; metric: saved prefill ms per resident GiB-s. Not run.
C1	Exact stochastic block verification	CPU · REAL CODE	OPT-IN	GPU A/B at temperature 0.7, PLD-free, B1 and B4 chain: accepted tokens per verify and tok/s. Not run.
C2	Routed A8 Gate+Up	M5 GPU · MEASURED	DEFERRED	Not scheduled; re-derive the bound on the post-B1' profile first. W8A8 does not apply to the 27B.
C3	Conditioned GDN block algebra	M5 GPU · MEASURED	REJECTED	None. chunk16 stays rejected (inexact for $\leq 1\%$).

ITEM	WORK PACKAGE	EVIDENCE	VERDICT	NEXT STEP
D1	Output-aware KV and directional recurrent precision	REPOSITORY-RECORDED	PENDING	Not scheduled; behind the open items in chapter 11.
D2	Learned capsule composition / HySparse2	PROPOSED	DEFERRED	Persona and routing capsules on the 9B, offline, with the Part B controls.

11.1 The campaigns, and the runs still missing

CAMPAIGN · TREE	WHAT IT SETTLES	RESULT	STATUS
estimator INTERMEDIATE	A2 on the GPU: cost_model vs running_max on identical code, 27B native MTP and ordinary, 12K contention and B4, 2 reps	cost model: neighbour gap p95 –15% (MTP) / –23% (ordinary), max –13% / –18%; long TTFT +6% / +1%; B4 unchanged	LANDED
post-q38-routes INTERMEDIATE	Four 27B routes after the first fixes (intermediate tree)	MTP B4 fixed, PLD warm reuse fixed; external-draft and PLD stalls unchanged → rfix-sched	LANDED
post-fn INTERMEDIATE	Flash-Next after the first fixes (intermediate tree)	stall 3.8 s, not solved; co-arrived prompts admitted in waves → rfix-fn	LANDED
thermal-q38 BASELINE	Thermal map at equal heat: ordinary, native MTP, DFlash2	≈ 90 s burst at 89 W, then ≈ 40 W budget; pacing costs 17-24% J/token	LANDED
keepwarm INTERMEDIATE	Idle-gap TTFT at 2, 6 and 20 s, keep-warm off vs on, 3 reps × 3 requests	wake penalty ≈ 0.3 s at 20 s idle; keep opt-in	LANDED
telemetry-overhead INTERMEDIATE	Decode tok/s with --power-telemetry off vs on, 3 reps	no measurable overhead	LANDED
governor INTERMEDIATE	Governor modes, four servers loaded, strict rotation (intermediate tree)	budget holds the cap; efficient 2-4% lower J/token; idle windup suspected → anti-windup added	LANDED
final-governor FINAL	max_throughput vs budget 25 W after anti-windup	25 W cap held at –59% power, –67% decode; B1 throttling is correct (B1 draws ≈ 37 W GPU+DRAM)	LANDED
final-q38-routes FINAL	Four 27B routes on the final tree	every route's neighbour p95 424-508 ms; MTP B4 1.5× ordinary; PLD B4 at parity	LANDED
final-fn FINAL	Flash-Next served vs ordinary on the final tree	served stall 6.9-9.5 s → 0.44-0.47 s; B8 same throughput, median TTFT halved	LANDED
final-q36-6bit FINAL	35B Heretic, NAX widths 4/8 vs 4/6/8, 3 reps	8K TTFT –7%, 16K –8%, B8 +4.5% (wall time), bit-identical	LANDED

CAMPAIGN · TREE	WHAT IT SETTLES	RESULT	STATUS
df flash-comp FINAL	Default vs proposal_composition false on the pinned 27B tree route	equivalent after the fix: composition no longer attaches to tree routes	LANDED
GVR loaded-model A/B	Flash-Next 32K-128K with P99 selection time	decides B2	NOT RUN
Block verification A/B	Temperature 0.7, PLD-free, B1 and B4 chain route	decides C1	NOT RUN
Value retention trace	Multi-session agent trace at equal resident bytes	decides B3	NOT RUN
PLD plain-round fix	Plain rounds on the plain decode path, then B4 rerun	decides 4.2	NOT RUN
Requalification	Changed defaults and routes (below)	decides A2 B1' A1	NOT RUN

Baseline = origin/main 187c6aa23; intermediate = integration tree before the review fixes, used only where arms share identical code; final = main. Before/after numbers in this review compare baseline with final.

11.2 What integration found

FINDING	WHAT HAPPENED	STATUS
Default DFlash2 route would not start	Main commit 66b2f9d1e defaulted proposal composition on for every external drafter with draft distributions; the pinned 27B pair runs a tree, where composition is refused, so the server failed at startup. Caught only because the final A/B drove the default route; the composition A/B then showed the tree route unaffected (section 4.5).	FIXED [S29] [E19]
Engine import-order regression	Validating <code>apc_retention_policy</code> imported <code>runtime.apc_v2</code> , which loads <code>runtime.models.base</code> before any adapter pins its import-time environment, so every guarded adapter (27B, Flash-Next, ...) refused to start with <code>ImportOrderError</code> . The parser moved to a dependency-free module, with a fresh-interpreter test.	FIXED [S30]
COW-1 ledger forced a device sync per store	The shared-snapshot ledger keyed by data address, which evaluates the array, so every APCv2 store during prefill waited on the GPU. The intermediate campaigns carry this cost; the final ones do not.	FIXED [S34] [E09]
Review fixes (codex review and the review lanes)	PLD rollback on a failed hybrid round (S37); strict acceptance-uniform comparison in block and self-MTP verification (S36); law validation before the greedy shortcut (S40); complete GPU + DRAM readings only for the governor, checked native sampler results, serialised sampling, a time-bounded attribution window (S38); exact withdrawal of refused APCv2 hits and freeable-only unleased bytes (S39).	FIXED [S36] [S37] [S38] [S39] [S40]
Two corrected readings	The B8 Flash-Next "regression" was a metric artifact (batch wall time is the measure); the governor's B1 "windup" was correct throttling, because B1 decode draws about 37 W of GPU + DRAM (chapter 10).	FIXED · TEXT [E09] [E17] [E23]

11.3 Open limits

- **PLD B4 reached parity late in the sweep:** 55–63 tok/s against ordinary 62–63 on the closing run (was 25–33), after all-token polling, cached batched views and pipelined plain rounds (82bdcc230; final-pld evidence). B1 still costs about 10% over ordinary.
- **Native MTP warm hit depth is unstable** (4,096 or 7,168 tokens restored on the same workload), and its contention max gap still has outliers up to about 1 s. [E15]
- **DFlash2 warm reuse is limited by design** (paired checkpoints only at the prompt boundary). [S22]
- **Test suite:** the full CPU suite passes on the M5 at the final tree (11,646 tests, 42/42 import guards); remaining M3 failures are environmental (no mlx_vlm, native binaries or TensorFlow checkout). [E09]
- **Requalification is needed** for every changed default or route: the cost-model estimator (now part of route identity), 6-bit NAX admission, the stall-bounded external routes, composition on chain drafters, and the harness re-pins. Adopted is not qualified.

11.4 v2 plan, ordered by measured prize

1. **Requalify what changed by default**, then fix PLD plain rounds and rerun B4. These are the only open items on default routes.
2. **Warm reuse:** stabilise native MTP's hit depth; price paired target + draft checkpoints for DFlash2 against the agent workloads that need them.
3. **Equal-heat methodology everywhere.** Every quoted A/B uses alternation and reports power and J/token next to tok/s.
4. **Governor next steps:** committed ramp reporting, a per-route operating curve from the thermal map, then work shifting. The cap itself is measured.
5. **Shared-prefix in-flight wait** (peer gufo #382): followers wait for a leader's shared prefix instead of prefilling it twice. Not built.
6. **Opt-in candidates to decide:** GVR (loaded A/B), block verification (sampled A/B), value retention (trace A/B); then routed A8 on the post-6-bit profile, and the research bets (D1, D2) with controls defined in advance.

CHAPTER 12

Peer work, revisited

v1 selected sixteen pull requests with a concrete MLX2 seam. Here is what became of each idea, and three more the sweep's peer survey added.

The sweep's peer survey pulled merged pull requests since 22 September from oMLX, Rapid-MLX, llama.cpp, vLLM, SGLang, TensorRT-LLM and others, then grepped MLX2 source and provenance before calling anything missing. Most of what peers shipped in the window MLX2 already had: context-copy drafts, eager async-eval dispatch, decode-end checkpoints, float32 top-p mass, EOS truncation inside speculative commits, LoRA fingerprints in the cache scope. Three peer bugs applied and were fixed (Anthropic tool-search blocks, lone UTF-16 surrogates returning 500, a dropped leading space on raw completions). [\[E10\]](#)

ID	PEER WORK	STATUS AT V1	V2 OUTCOME FOR MLX2
P01	FlashInfer #4106: distribution-aware MoE autotuning	merged 13 Aug	B1: measured not worth building on M5 (E05); 6-bit NAX built instead.
P02	TensorRT-LLM #18702: self-sampling GVR V2 prefill top-k	merged 9 Sep	B2: ported as S12, merged opt-in.
P03	TensorRT-LLM #19076: device-aware GVR V2 dispatch	merged 16 Sep	Crossover must be re-measured on Metal.
P04	oMLX #4320: routed A8 Gate/Up prefill	open 6 Oct	C2: Amdahl bound E06.
P05	oMLX #4324: small-batch GLM router changes	open 6 Oct	No mlx2 GLM adapter; watch only.
P06	oMLX #4312: long-context cache-copy and join fixes	merged 6 Oct	Covered at the v1 pin.
P07	oMLX #4314: no duplicate paged KV in completion snapshots	merged 6 Oct	mlx2 skips stores on native paged routes.
P08	RapidMLX #4238: pipeline parked MTP rounds	merged 6 Oct	Handoff/overlap audit; not done.
P09	MLX #4633: configurable Metal residency refresh	merged 6 Oct	Related to keep-warm (S07); read only.
P10	mlx-lm #1929: bounded-memory quantized attention tiling	open 27 Sep	Read only; qSDPA row tiling already present.
P11	llama.cpp #30065: few-row Metal MMA routes	draft 6 Oct	Lane matmul covers small-M.
P12	vLLM #60083: cache per-request sparse-page residency	merged 6 Oct	No matching mlx2 bottleneck found.
P13	vLLM #56318: cached prompt tokens by source tier	merged 6 Oct	APC-2 per-request hit accounting fixed (S06).
P14	SGLang #42297: opt-in stochastic block verification	merged 5 Oct	C1: motivation for S04 (not read).

ID	PEER WORK	STATUS AT V1	V2 OUTCOME FOR MLX2
P15	LMCache #5500: typed FP8 Re-RoPE cache support	merged 6 Oct	Typed representation metadata.
P16	Exo #2393: RDMA/stream-order fix	open 5 Oct	Distributed only.
P17	RapidMLX #4256: cost-model correction after pipelining	merged 6 Oct	Recalibrate costs after kernel changes.
P18	vLLM #58463: remove redundant MTP attention metadata	merged 6 Oct	Host work that overlaps is not on the critical path.
P19	llama.cpp #30042: shared latent reads for sparse GLM attention	merged 6 Oct	Consumer geometry matters.
P20	ik_llama.cpp #2600/#2601: tensor identity and compressed-state ownership	merged 6 Oct	Ownership invariants.
P21	SGLang #42651: coordinate draft and target KV budgets	merged 6 Oct	Combined memory invariant.
P22	LMCache #5239/#4164: cache admission and SSD recovery	merged 5 Oct	Admission policy references.
P23	oMLX #4112, RapidMLX #4221/#4075: speculative controller status	6 Oct	Park-memory and stable-greedy invariants already adopted.
P24	oMLX #3974: GPU keep-warm ticker	merged	Peer claim: ~1.5 s idle then up to ~1.7 s stall; built as S07; measured ≈ 0.3 s on the M5 Max (E21).
P25	gufo #382: shared-prefix in-flight wait	merged	Followers wait for a leader's shared prefix; not built (chapter 11, plan item 5).
P26	vLLM #57693 / oMLX #4201: Anthropic tool-search blocks	merged	Fixed in the sweep (fc7ca8777).

Upstream policy: nothing in this review was filed with ml-explore projects; MLX and mlx-lm pull requests are read-only references.

CHAPTER 13

Method, limits and what v2 can conclude

v1 listed five outcomes to measure. v2 adds a sixth, and rules about heat, defaults and batch metrics.

Six outcomes

- **User-visible latency:** TTFT, inter-token gaps (the neighbour max gap is now a first-class column), completion time.
- **Useful throughput:** committed target tokens per second; physical rows and target passes as explanation only.
- **Memory and I/O:** unique resident bytes (shared checkpoints counted once), peak Metal bytes, swap.

- **Numerical and task behaviour:** the route's actual exactness gate, first divergence and margin, near-tie rate per thousand tokens.
- **Operational robustness:** cancellation, partial hits, admission declines, failed draft paths.
- **Energy and heat (new):** GPU + DRAM joules per output token, mean P-state, die temperature, measured at equal heat.

Rules the sweep added

1. **Alternate arms at equal heat** for any number that will be quoted; never compare a cool first arm with hot later ones.
2. **Name the falsifier in advance.** B1 set a 3% bar before measuring and stopped at 0.5%; C3 set the prize before the kernel.
3. **Measure the fraction before the kernel** (Figure 5.2). A 1.4x kernel on 1% of the request is 0.3%.
4. **Count what serving uses, not what a gate measured:** CAP-2 passed a mechanism gate on a seam serving did not use.
5. **Default-on means required:** a selected mechanism that never runs must fail qualification (S18).
6. **Drive the default route end to end before calling a merge done:** the composition default broke the main 27B route and only the final A/B noticed (S29).
7. **Measure batches by wall time:** an aggregate rate taken from each request's first token rewards staggered starts.

What v2 can and cannot conclude

The GPU evidence is A/B evidence: two or three rotated reps on one M5 Max through the real serving path, swap-free, not qualification and not controlled performance runs. Baseline and final trees differ in many commits, so a before/after delta says what the sweep as a whole changed; only the intermediate A/Bs whose arms share identical code (estimator, keep-warm, telemetry, governor) isolate one mechanism. That is enough to reject the B1 planner and C3, to adopt the cost-model estimator and 6-bit NAX as defaults pending requalification, and to show which serving seams cost the most and that they are fixed. It is not enough to quote speedups as product numbers, which need controlled runs. CPU evidence is exact about logic (oracles, policy simulators, serving repros) and silent about time. Peer and paper results keep their authors' hardware and workload scope. Where a lane report or the coordinator's ledger is the only source (restored-token counts at the baseline, the retracted claims, PLD cycle counts, the overflow note in C3), the citation says so.

THE JUDGMENT, UPDATED

v1's central claim holds: MLX2 gains most from making its advanced paths measurable, numerically coherent and cheap to combine. The GPU adds three corrections. The biggest losses were in scheduling and serving seams, which only a whole-request measurement through the server could expose; the final runs show most of them closed. The one kernel win was a refused bit width, found by measuring a fraction before building a kernel. And on a laptop GPU, heat and energy are part of the result: the GPU runs on a power budget, a cap on that budget is now a measured control, and numbers should be taken at equal heat and reported in joules as well as tokens per second.

CHAPTER 14

Source register

v1 identifiers are kept so that its citations still resolve. Code links point at the v1 pin on the public mirror; the sweep's code is public as 13 signed topic commits (pierre427/mlx2 008eb01..8359344), and its commits and evidence live in the private tree (main 43d561e38).

Sweep commits (v2)

[S01] d28a774aa **Bound prefill stalls with a completion-time cost model, MTP preparation included**

A2: cost_model estimator, default; GPU A/B in E16.

[S02] 56fce148e **Hold co-arrived self-MTP prefills so multi-chunk prompts form one cohort**

SS-1: the MTP B4 cohort lock.

[S03] 2942ee30b **Batch prompt lookup on hybrid recurrent targets**

SPEC-10: PLD B4 serial on Qwen3.8.

[S04] ccbf2a99c **Add opt-in exact block verification for the linear external chain**

C1: block verification, opt-in.

[S05] f44f355b5 **Add opt-in value-aware APCv2 retention**

B3: value retention, opt-in.

[S06] cd9fb13b3 **Count APCv2 hits per request and withdraw refused hits**

APC-2, prerequisite of B3.

[S07] 726837c12 **Add opt-in GPU keep-warm ticker for idle gaps between requests**

Keep-warm, opt-in.

[S08] e5514a5d0 **Keep retained trajectory waypoints exact and report honest errors**

A3 / CAP-1.

[S09] cd330ca82 **Apply served semantic memory on the prompt-tail forward**

CAP-2 (+ CAP-3, CAP-7).

[S10] a809a970d **Publish opt-in power and thermal telemetry from the server**

Energy telemetry; governor input.

[S11] 01c33b86f **Add unprivileged power/DVFS/temperature telemetry and instrumented profilers**

profile_features.py and thermal_map.py.

[S12] 41b87aa11 **Add GVR-style self-sampling exact QSA selector candidate (default off)**

B2 (branch claude/qa-gvr-1006, merged opt-in as S31).

[S13] 4a7ee991a **Research: MoE NAX tile headroom, A8 gate+up Amdahl, GDN chunked prefill microbench**

B1, C2, C3.

[S14] cc4e96bac **Charge shared APCv2 checkpoint snapshots once**

COW-1, the B3 denominator.

[S15] 026058e1b **Let capsule requests restore the exact ordinary prefix**

CAP-6.

[S16] fb21b8a41 **Prelaunch tree drafts only for singleton cohorts and count discards**

SPEC-01.

[S17] 883c0b638 **Verify greedy external chains by token compare**

SPEC-06.

[S18] 734a79679 **Require engagement of the default-on mechanisms the qualifier skipped**

CQ-4 false-greens.

[S19] 29056b2fb **Treat the external varlen pair as one operator-owned default group**

A1 / varlen flags.

[S20] d23b16895 **Cut warm continuations at the cold prefill plan's boundaries**

APC-3.

[S21] f6a160933 **Serve the deepest sidecar ancestor on draft routes**

APC-1.

[S22] 583d78bcd **Document shared-prefix reuse on the speculative routes**

Warm-reuse limit of external draft.

[S23] 0182ea9e7 **Add an idle-gap TTFT workload to the feature profiler**

Keep-warm A/B workload.

[S24] 48289d962 **Let adapters declare the prefill row tile used by the stall grid and mixed forwards**

SS-7 row tile.

[S25] 46656a8ec **NAX sorted gather: unpack affine 2/3/5/6-bit, route admission by width**

B1': the 6-bit kernel.

[S26] 7e28b1ace **NAX sorted gather: admit affine 6-bit by default; provenance**

B1': default widths 4/6/8.

[S27] 071fbb591 **Wire the power governor into serving with an admin control endpoint (controller 6dc3cfb03)**

Governor, default off.

[S28] f030c22fa **Keep the budget governor from winding up while its engine is idle**

Anti-windup for shared hosts.

[S29] e33356767 **Keep default proposal composition off tree routes so DFlash2 starts again**

Fixes main 66b2f9d1e on the default route.

[S30] 9f9a81f86 **Parse the APC retention policy without importing model modules**

Engine import-order regression.

[S31] a768539bf **Merge claude/qsa-gvr-1006 (bench 223068567)**

B2 merged opt-in.

[S32] 1c45a38a7 **Keep the B4 prompt-lookup cohort whole and drop per-row host syncs**

PLD B4 fragmentation.

[S33] 673aa9fe0 **Stall-bound external-draft and prompt-lookup prefill beside decode**

Fairness for the two routes that had none.

[S34] b02360126 **Key the APCv2 snapshot ledger without evaluating arrays**

COW-1 ledger device sync.

[S35] 057373692 **Prepare co-arrived self-MTP prompts together after a partial first boundary (+ e4213207f)**

Flash-Next B8 waves; per-sibling hold.

[S36] 8ca62ebf5 **Compare acceptance uniforms strictly in block and self-MTP verification**

Review fix (C1).

[S37] 1d25dd7d1 **Rewind recurrent rows on hybrid abort and rebuild PLD lanes on failed rounds**

Review fix (PLD rollback).

[S38] dffad06da **Feed the power governor only complete GPU + DRAM readings (+ f52578e61, f03a51b6e, 01fd7a3a2)**

Review fixes (telemetry and governor safety).

[S39] 73548bdba **Withdraw a refused APCv2 hit's served depth and reuse count exactly (+ b310c3e13)**

Review fixes (APC accounting).

[S40] 91073ecad **Validate proposal laws fully before the greedy verify shortcut**

Review fix (S17 shortcut).

[S41] 470e7d3af **Charge each self-MTP one-call preparation its own fixed cost and depth**

Burst preparation stall bound.

[S42] 8c7e1fe78 **Expire stale cost-model samples when a stall bound is asked for**

Cost-model ageing.

Evidence files (v2)

[E01] qualification/runs/sweep-20261006/q38-routes/

Qwen3.8-27B four-route GPU campaign (summary, results, spec).

[E02] qualification/runs/sweep-20261006/fn/

Flash-Next served vs ordinary vs QSA direct4.

[E03] qualification/runs/sweep-20261006/q38-features/

27B interior checkpoints, lane matmul, energy per rep.

[E04] qualification/runs/sweep-20261006/q36/

Qwen3.6-35B-A3B ordinary.

[E05] qualification/runs/research-20261006/b1-moe-tile-headroom/

B1: routing capture, 36-plan grid, analysis (README, analysis-tiles-r1.json).

[E06] qualification/runs/research-20261006/c2-a8-gate-up-amdahl/README.md

C2 Amdahl bound.

[E07] qualification/runs/research-20261006/c3-gdn-prefill/

C3 GDN microbenchmark (README, gdn.json).

[E08] qualification/runs/qsa-stage1-gvr-20261006/bench.json

B2 GVR benchmark at 0a1c483ca: 9 selector, 9 stage-one and 9 adversarial cells, synthetic inputs, M5 Max (commit 223068567; the earlier 6-cell run is research-20261006/b2-qsa-gvr/).

[E09] qualification/runs/sweep-20261006/FINDINGS-GPU.md

Coordinator's GPU ledger: authoritative interpretations, the retracted and corrected claims, q38-features2, PLD cycle counts.

[E10] qualification/runs/sweep-20261006/lanes/*.md

Lane reports: serving-sched (SS-*), spec (SPEC-*), capsules (CAP-*), apc-state (APC-*), COW-*), kernels-rows (KR-*), config-qual (CQ-*), pr-survey, history.

[E11] docs/research-review/data/cpu-sims.json

This review's CPU runs of real code: estimator trace, block-verification alignment sweep, routing histogram (gen_cpu_data.py).

[E12] `provenance/block-verification.json`

Block verification provenance and validation record.

[E13] `qualification/runs/sweep-20261006/thermal-q38/`

Thermal map at equal heat (baseline-era tree 38129cc92).

[E14] `qualification/runs/sweep-20261006/post-q38-routes/`

27B four routes on the intermediate tree (816aef02c, rep 0 + partial rep 1).

[E15] `qualification/runs/sweep-20261006/final-q38-routes/`

27B four routes on the final tree, 2 reps.

[E16] `qualification/runs/sweep-20261006/estimator/`

cost_model vs running_max on identical code, 27B MTP and ordinary, 2 reps.

[E17] `qualification/runs/sweep-20261006/final-fn/`

Flash-Next served vs ordinary on the final tree (post-fn/ holds the intermediate run).

[E18] `qualification/runs/sweep-20261006/final-q36-6bit/`

35B Heretic, NAX widths 4/8 vs 4/6/8, final tree, 3 reps.

[E19] `qualification/runs/sweep-20261006/dflash-comp/`

Default vs proposal_composition false on the pinned 27B tree route.

[E20] `qualification/runs/research-20261006/nax-6bit/`

6-bit NAX canaries, falsifier, per-call microbenchmark, adapter-level 8K ABBA A/B.

[E21] `qualification/runs/sweep-20261006/keepwarm/`

Idle-gap TTFT, keep-warm off vs on, 3 reps.

[E22] `qualification/runs/sweep-20261006/telemetry-overhead/`

Decode with power telemetry off vs on, 3 reps.

[E23] `qualification/runs/sweep-20261006/governor/, final-governor/`

Governor modes in strict rotation (run 1, intermediate tree) and max_throughput vs budget 25 W after anti-windup (final tree).

Pinned implementation (v1)

[C01] Architecture, capabilities and admission contracts

The implemented / qualified / selected / observed-used distinction and the route receipts that enforce it. v1 pin 008eb011fe49.

[Working rules](#); [Runtime architecture](#); [Capability contracts](#); [Capability routing](#); [Batch admission](#); [State manifests](#)

[C02] Artifact-bound Qwen3.8 DFlash defaults

The exact target/draft pair selects the host-pairwise tree15 policy with both varlen flags off. v1 pin 008eb011fe49.

[Qwen3.8 27B adapter](#); [Registry binding](#); [Owned TensorFold binding](#); [Vendored executor provenance](#)

[C03] TensorFold numerical-law correction and historical varlen evidence

Fused GDN arithmetic correction, unified tree GDN, and the 5 October varlen closeout as history. v1 pin 008eb011fe49.

[Fused GDN provenance](#); [Unified tree GDN provenance](#); [Tree GDN implementation](#); [Coupled-law host tests](#); [Tree GDN Metal parity tests](#); [Varlen DFlash closeout](#); [Varlen scheduler provenance](#)

[C04] Adaptive scheduling and prefill planning

DecodeTimeFairness, the stall bound and the prefill plan (v1 pin; v2 changes in S01). v1 pin 008eb011fe49.

[Adaptive policy](#); [Prefill plan](#); [Policy tests](#); [Peer scheduling provenance](#)

[C05] Paged packed scheduling and cost model

The research native route's deadline scheduler and its exact-shape price table. v1 pin 008eb011fe49.

[Pack scheduler](#); [Pack price](#); [Serving profile](#); [Native contract](#)

[C06] Speculative sampling and external drafting

Acceptance laws, proposal generation and verification on the external-draft route. v1 pin 008eb011fe49.

[Sampling laws](#); [External speculative](#); [Compact sampling tests](#); [DFlash2 host tests](#); [Tree bridge tests](#)

[C07] Exact-prefix cascade and serving refusal

The cascade opener refuses until the full sampler/RNG/history handoff is owned. v1 pin 008eb011fe49.

[Serving bridge](#); [Cascade](#); [State binding](#); [Host tests](#); [Serving integration](#)

[C08] MoE NAX gather and prefill eligibility

The indexed expert kernel, its mean-based _plan and the 4/8-bit support gate. v1 pin 008eb011fe49.

[NAX gather](#); [NAX tests](#); [Prefill-only tests](#)

[C09] Dense W8A8 prefill and weight residency

Instance-scoped approximate W8A8 prefill and process-lifetime weight residency. v1 pin 008eb011fe49.

[W8A8 prefill](#); [W8A8 tests](#); [Weight residency](#); [Serving lifecycle](#)

[C10] QSA selector, stage-one scoring and segmented readers

Exact radix selection, direct4/direct8 candidates and the score/reader split. v1 pin 008eb011fe49.

[QSA stage one](#); [QSA selector](#); [QSA scores](#); [Indexed reader](#); [Segmented Metal](#); [Segmented reference](#)

[C11] Gated delta state, fused prefill and recurrent quantization

GDN update, chunked prefill scan and the state-precision research records. v1 pin 008eb011fe49.

[Gated delta](#); [GDN state](#); [Fused GDN prefill](#); [Qwen3.5 caches](#); [State quant research](#); [FP16 state](#)

[C12] KV cache quantization, direct decode and verification

Direct compressed attention and the KV fidelity harness. v1 pin 008eb011fe49.

[Cache](#); [Model base](#); [qSDPA decode](#); [qSDPA verify](#); [KV fidelity](#)

[C13] APCv2 placement and exact state binding

Radix cache, interior placement, native restore and state identity. v1 pin 008eb011fe49.

[APCv2](#); [Interior placement](#); [State binding](#); [Native restore](#); [State manifests](#); [Placement tests](#)

[C14] Activation capsule compilation and reconstruction

The trajectory compiler whose metric v1 probed (fixed in S08). v1 pin 008eb011fe49.

[Capsule compiler](#); [Capsule tests](#); [Semantic capsules](#)

[C15] Capsule injection, composition and serving hooks

Bridge binding, the bus, the 9B adapter hook and serving admission. v1 pin 008eb011fe49.

[Activation injection](#); [Capsule bus](#); [Qwen3.5 9B adapter](#); [Hidden-state hook](#); [Serving admission](#); [Capsule serving tests](#)

[C16] Neural concepts and semantic memory

Concept payloads and the adapter scheduling that applies them. v1 pin 008eb011fe49.

[Neural concepts](#); [Concept tests](#); [Semantic memory](#); [Adapter scheduling](#)

[C17] Experimental HySparse2 and own-layer quantized key indexes

A trained research architecture and a selection index that keeps full KV. v1 pin 008eb011fe49.

[HySparse2 model](#); [HySparse2 attention](#); [HySparse2 config](#); [HySparse2 APC](#); [Key index](#); [Key index tests](#)

[C18] Spomin standard and live cache surgery

Token removal and position repair, with their approximation bounds. v1 pin 008eb011fe49.

[Standard surgery](#); [Live surgery](#); [Layer surgery](#); [Surgery tests](#); [Live provenance](#)

[C19] Tokenizer worker, API and adapter boundary

Tokenizer isolation, the serving API and the backend-neutral adapter contract. v1 pin 008eb011fe49.

[Tokenizer worker](#); [Worker tests](#); [Serving API](#); [Request ownership](#); [Adapter contract](#); [Tokenizer integrity](#)

[C20] Historical wins, regressions and closed experiments

Repository-recorded negatives that constrain new proposals. v1 pin 008eb011fe49.

[ASPIRE closeout](#); [ASPIRE provenance](#); [MoE weighted reduction](#); [Batched GDN verify](#); [TensorFold q4 QMV](#); [N20 MLP/GDN factorial](#); [QSA stage-one candidate](#); [MoE rhs pad](#); [TensorFold long context](#)

[C21] Metal prefill generators, vendored QMM and cooperative QSA

Where the Metal tensor-op tiling and eligibility live. v1 pin 008eb011fe49.

[Prefill generator](#); [Lane QMM](#); [NAX QSA](#); [TensorFold prefill](#); [Long paged NAX](#)

[C22] 6 October peer-derived serving invariants

Output budgets, APC joins, template sandboxing and deferred admission. v1 pin 008eb011fe49.

[Output-budget provenance](#); [Scheduling provenance](#); [Output-budget tests](#); [APC join tests](#); [Template sandbox tests](#); [Deferred admission tests](#)

Primary research and documentation (v1)

[R01] Evaluating the Applicability of LLM Inference Optimizations on Apple Silicon

MacPaw Research, 30 Sep 2026. Apple-specific evidence that gains are workload- and device-bound. [MacPaw](#)

[R02] SLOWeave: Deadline-Aware Adaptive Prefill Chunking

preprint, 7 Sep 2026. Iteration-cost prediction against decode deadlines; the shape of S01. [arXiv:2609.07883](#)

[R03] DLFP: prefill timing and scheduling (descriptive label)

preprint, 29 Sep 2026. Completed-time measurement over scheduling-interval proxies. [arXiv:2609.38386](#)

[R04] CaDDTree: Cost-Aware Diffusion Draft Trees

preprint, 1 Jun 2026. Joint tree shape and node budget under verify cost. [arXiv:2606.01813](#)

[R05] LibraSpec: Marginal-Gain-Driven Diffusion Speculative Decoding

preprint, 9 Aug 2026. How many drafted tokens are worth verifying. [arXiv:2608.08721](#)

[R06] Block Verification Accelerates Speculative Decoding

Sun et al., 2024. Algorithm 2 is what S04 implements. [arXiv:2403.10444](#); [SGLang speculative docs](#)

[R07] DA-MoE: Distribution-Aware MoE Inference with Adaptive Kernel Dispatch

preprint, Jul-Aug 2026. Routing-skew-aware dispatch; v2 measured the M5 headroom (E05). [arXiv:2607.23099](#)

[R08] GVR V2: Self-Sampling and Multi-Thresholding for Exact Top-K

NVIDIA TensorRT-LLM blog, Sep 2026. The selector law S12 ports to Metal. [NVIDIA blog](#)

[R09] Optimize custom ML operations with Metal tensors

Apple WWDC26 session 330. TensorOps and cooperative layouts behind NAX kernels. [WWDC26 330](#)

[R10] Gated DeltaNet block reformulation research

Mar 2026; FLA #797. Block algebra v1 proposed; E07 measured the chunked alternative. [Veitner](#); [FLA #797](#); [Alcaide](#)

[R11] DAMP: Decay-Aware Mixed-Precision Recurrent-State Quantization

preprint, Aug-Sep 2026. Recurrent-state precision; D1 background.
[arXiv:2608.27513](#)

[R12] OptR: Output-Aware Rotation for INT2 KV-Cache Quantization

preprint v4, 23 Sep 2026. Output-aware KV objective (D1).
[arXiv:2608.02691v4](#)

[R13] OSCAR: Spectral Covariance-Aware Rotation for 2-bit KV

preprint, 18 May 2026. Offline attention-aware calibration (D1).
[arXiv:2605.17757](#)

[R14] Marconi: Prefix Caching for the Era of Hybrid LLMs

MLSys 2025. Saved compute per byte; the value function of S05.
[arXiv:2411.19379](#)

[R15] Janus: Agentic Serving over SSD-based Sparse KV

preprint, 29 Sep 2026. Prefetch prediction with exact miss recovery.
[arXiv:2609.36938](#)

[R16] RelaxKV: Query-Guided Recomputation for KV Reuse

preprint, 27 Sep 2026. Declared-approximate cache repair.
[arXiv:2609.33503](#)

[R17] Cartridges++: KV Compression without Off-Context Derailment

preprint, 28 Sep 2026. Off-context controls for learned memory (D2).
[arXiv:2609.35621](#)

[R18] Cartridges at Scale: Modular KV Caches over Document Collections

preprint, 3 Jun 2026. Composition with distractors (D2).
[arXiv:2606.04557](#)

[R19] Theseus: Transporting Task Vectors across Architectures

ICML 2026. Functional alignment before cross-model transfer.
[arXiv:2602.12952](#); [PMLR](#)

[R20] HySparse2: Hybrid Sparse Attention with Two-Level KV Sharing

preprint, 22 Sep 2026. Architecture-level FLOP claims, not M5 wall clock.
[arXiv:2609.26368](#)

[R21] EcoSpec

preprint, Jul 2026. Expert-traffic view of speculative blocks.
[arXiv:2607.12696](#)

[R22] BaseRT: native Metal inference and M5 neural-accelerator research

Jul 2026. Prefill and decode as different resource problems.
[arXiv:2607.19438](#); [arXiv:2607.00501](#)

[R23] NPUMoE: MoE Inference with Apple Silicon NPUs

preprint, 20 Apr 2026. ANE expert offload; transfer costs decide.
[arXiv:2604.18788](#)

[R24] SonicMoE: IO and Tile-aware MoE

ICLR 2026. Tile-aware MoE design.
[ICLR 2026](#)

[R25] TensorRT-LLM cache compression and job-level evaluation

NVIDIA docs, Sep 2026. Job-level metrics over token-rate metrics.
[KV compression](#); [Agentic trace replay](#)

[R26] SGLang HiCache and session-aware radix caching

SGLang docs, Oct 2026. Hierarchical and session-aware retention.
[HiCache](#); [Session radix](#)

[R27] TurboQuant: Online Vector Quantization

ICLR 2026. Vector distortion vs inner-product bias.
[arXiv:2504.19874](#)

[R28] ParallelFlow: Parallelizing Linear Transformers via Flow Discretization

preprint, Apr 2025. Parallel recurrence theory; no Metal claim.
[arXiv:2504.00492](#)

[R29] CompKV: Compensation-Aware KV Selection

preprint, 22 Sep 2026. Approximate selection with residual compensation.
[arXiv:2609.26300](#)

Peer pull requests (v1, plus P24-P26)**[P01] FlashInfer #4106: distribution-aware MoE autotuning**

merged 13 Aug. B1: measured not worth building on M5 (E05); 6-bit NAX built instead.
[flashinfer #4106](#)

[P02] TensorRT-LLM #18702: self-sampling GVR V2 prefill top-k

merged 9 Sep. B2: ported as S12, merged opt-in.
[TRT-LLM #18702](#)

[P03] TensorRT-LLM #19076: device-aware GVR V2 dispatch

merged 16 Sep. Crossover must be re-measured on Metal.
[TRT-LLM #19076](#)

[P04] oMLX #4320: routed A8 Gate/Up prefill

open 6 Oct. C2: Amdahl bound E06.
[omlx #4320](#)

[P05] oMLX #4324: small-batch GLM router changes

open 6 Oct. No mlx2 GLM adapter; watch only.
[omlx #4324](#)

[P06] oMLX #4312: long-context cache-copy and join fixes

merged 6 Oct. Covered at the v1 pin.
[omlx #4312](#)

[P07] oMLX #4314: no duplicate paged KV in completion snapshots

merged 6 Oct. mlx2 skips stores on native paged routes.

[omlx #4314](#)**[P08] RapidMLX #4238: pipeline parked MTP rounds**

merged 6 Oct. Handoff/overlap audit; not done.

[Rapid-MLX #4238](#)**[P09] MLX #4633: configurable Metal residency refresh**

merged 6 Oct. Related to keep-warm (S07); read only.

[mlx #4633](#)**[P10] mlx-lm #1929: bounded-memory quantized attention tiling**

open 27 Sep. Read only; qSDPA row tiling already present.

[mlx-lm #1929](#)**[P11] llama.cpp #30065: few-row Metal MMA routes**

draft 6 Oct. Lane matmul covers small-M.

[llama.cpp #30065](#)**[P12] vLLM #60083: cache per-request sparse-page residency**

merged 6 Oct. No matching mlx2 bottleneck found.

[vLLM #60083](#)**[P13] vLLM #56318: cached prompt tokens by source tier**

merged 6 Oct. APC-2 per-request hit accounting fixed (S06).

[vLLM #56318](#)**[P14] SGLang #42297: opt-in stochastic block verification**

merged 5 Oct. C1: motivation for S04 (not read).

[SGLang #42297](#)**[P15] LMCache #5500: typed FP8 Re-RoPE cache support**

merged 6 Oct. Typed representation metadata.

[LMCache #5500](#)**[P16] Exo #2393: RDMA/stream-order fix**

open 5 Oct. Distributed only.

[exo #2393](#)**[P17] RapidMLX #4256: cost-model correction after pipelining**

merged 6 Oct. Recalibrate costs after kernel changes.

[Rapid-MLX #4256](#)**[P18] vLLM #58463: remove redundant MTP attention metadata**

merged 6 Oct. Host work that overlaps is not on the critical path.

[vLLM #58463](#)**[P19] llama.cpp #30042: shared latent reads for sparse GLM attention**

merged 6 Oct. Consumer geometry matters.

[llama.cpp #30042](#)**[P20] ik_llama.cpp #2600/#2601: tensor identity and compressed-state ownership**

merged 6 Oct. Ownership invariants.

[#2600; #2601](#)**[P21] SGLang #42651: coordinate draft and target KV budgets**

merged 6 Oct. Combined memory invariant.

[SGLang #42651](#)**[P22] LMCache #5239/#4164: cache admission and SSD recovery**

merged 5 Oct. Admission policy references.

[#5239; #4164](#)**[P23] oMLX #4112, RapidMLX #4221/#4075: speculative controller status**

6 Oct. Park-memory and stable-greedy invariants already adopted.

[omlx #4112; Rapid #4221; Rapid #4075](#)**[P24] oMLX #3974: GPU keep-warm ticker**

merged. Peer claim: ~1.5 s idle then up to ~1.7 s stall; built as S07; measured ≈ 0.3 s on the M5 Max (E21).

[omlx #3974](#)**[P25] gufo #382: shared-prefix in-flight wait**

merged. Followers wait for a leader's shared prefix; not built (chapter 11, plan item 5).

[P26] vLLM #57693 / oMLX #4201: Anthropic tool-search blocks

merged. Fixed in the sweep (fc7ca8777).

[vLLM #57693; omlx #4201](#)